

A Unified Approach for Assessing Sensitivity to Violations of Causal Assumptions*

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Abstract

This paper introduces a general method for sensitivity analysis to assess the robustness of causal estimates when key assumptions are violated. Unlike prior sensitivity approaches that are developed on a case-by-case basis, it presents a unified framework that accommodates a wide variety of assumption types, including (1) functional-form violations, e.g. the presence of defiers in studies that assume monotonicity; (2) exclusion-restriction violations, e.g. an encouragement that directly affects the outcome in instrumental-variable studies; and (3) unconfoundedness violations, e.g. unobserved common causes in selection-on-observables studies. The key innovation is to allow for an assumption to be violated in a proportion of the data that is at most θ ; at $\theta = 0$, the approach recovers the original estimates, and $\theta = 1$ is equivalent to discarding the assumption entirely. By using recent developments in partial identification, the method derives *sharp bounds* representing the range of possible conclusions for a given θ ; if the best- and worst-case conclusions both have the same sign (e.g., both positive), then the original estimate is said to be θ -robust. By varying θ , researchers can determine the severity of the violation needed to reverse their causal estimates, expressed in a form that is easy to reason about with domain expertise. The method’s flexibility is demonstrated with applications that reanalyze a number of published studies. For example, findings about habitual voting from [Davenport et al. \(2010\)](#)—the idea that voting in election t increases the chance of voting again in $t + 1$ —are 0.01-robust to violations of the exclusion restriction: if 1% of voter decisions at $t + 1$ may be directly influenced by a prior get-out-the-vote encouragement at t , conclusions about habit formation can no longer be supported. [Hobolt’s \(2016\)](#) finding that higher education leads to “stay” votes in the Brexit referendum is 0.06-robust to selection-on-observables violations: unobserved common causes of education and votes must be present in 6% of respondents to overturn results. Similarly, findings from [Knox et al. \(2020\)](#) on racially discriminatory police force are 0.06-robust to violations of a monotonic selection assumption: 6% of initial police decisions to stop must be discriminatory against white civilians before conclusions about subsequent use of force cannot be supported. This method offers a widely applicable, easy-to-interpret tool for examining the implications of assumption violations on causal conclusions.

Keywords: sensitivity analysis, causal inference, instrumental variables, partial identification

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Contents

1	Introduction	2
2	A Motivating Example	6
3	Formalizing the Proposed Sensitivity Analysis Approach	8
3.1	Notation for the Running IV Example	8
3.2	Notation and Theory for General Sensitivity Analyses	10
3.3	A Brief Primer on Principal Stratification and <code>autobounds</code>	13
3.4	Analysis of Relaxations for Three Assumption Types	16
3.5	Solving the sensitivity analysis problem	19
3.6	Statistical Inference	21
4	Worked Examples of Sensitivity Results for Common Designs	23
4.1	Instrumental Variables	23
4.2	Classic Confounding	27
4.3	Factorial Experiments	29
5	Empirical Results	32
5.1	Instrumental Variables: Reanalysis of Davenport et al. (2010)	32
5.2	Classic Confounding: Reanalysis of Hobolt (2016)	36
5.3	More Complex Scenarios: Reanalysis of Knox et al. (2020)	39
6	Conclusion	40
A	Principal Strata methods	45
A.1	Graph Stratification	45
A.2	Principal strata for instrumental variable models	46
B	Sensitivity Function Results	46
B.1	Symbolic sensitivity function for simple confounding	46
B.2	Symbolic sensitivity function for “no defiers” in instrumental variable models	48

1 Introduction

In recent decades, the quality of scientific results in the social sciences has improved due to a shift in focus toward better research designs and identification strategies (Angrist and Pischke, 2010). This shift, often referred to as the “credibility revolution,” emphasizes transparency in causal assumptions and the use of domain expertise to justify them. For example, social scientists now more frequently employ natural experiments, where assumptions about the absence of treatment-outcome confounding can be justified by expert knowledge, to analyze the causal effects of policies and other treatments. When these assumptions cannot be justified, many have argued for alternative methods like instrumental variables (IV; Imbens and Angrist, 1994; Angrist et al., 1996a; Angrist and Krueger, 2001) that encourage individuals to opt into treatment. However, IV analyses come with their own assumptions: even when randomization of the instrument is guaranteed by design, researchers must justify additional assumptions—including an *exclusion restriction* and *monotonicity*, defined more formally later—to draw conclusions about causal effects. More broadly, it is now widely recognized that most causal inferences rely on specific sets of assumptions that must be reasonably defended before conclusions can be drawn.

For many assumptions in applied research, debates about plausibility involve a false dichotomy: either an assumption is defensible or it is not. However, an important exception that motivates this work is the selection-on-observables approach, which relies on the assumption that there are no unobserved confounders that jointly determine treatment and outcomes. In this setting, sensitivity analyses are now widely used to assess the robustness of causal estimates—that is, the extent to which an assumption would need to be violated before the conclusion could no longer be supported. However, there are relatively fewer options for researchers that wish to probe key causal assumptions in other research designs: while a variety of sensitivity analyses have been developed on a case-by-case basis for specific methods and assumptions (Imbens, 2003; Imai et al., 2010; Blackwell, 2014; Knox et al., 2019; Keele et al., 2019), there are many more designs for which no such approach exists. In this paper, I propose a widely applicable approach that accommodates a broad range of assumption types under a single unified framework.

In the selection-on-observable setting, a wide variety of methods have been proposed to assess whether conclusions are robust to assumption violations (e.g. [Cornfield et al., 1959](#); [Rosenbaum and Rubin, 1983](#); [Imbens, 2003](#); [Brumback et al., 2004](#); [VanderWeele and Ding, 2017](#); [Cinelli and Hazlett, 2020](#); [Kennedy et al., 2019](#)). This approach proposed in this paper is most closely related to those that avoid restrictive parametric models and emphasize interpretability. An example of an easy-to-interpret but parametric sensitivity analyses is [Imbens \(2003\)](#), which treats unobserved confounders as a binary variable and models treatment and outcome with logistic and linear regressions. On the other hand, the popular “e-value” approach of [VanderWeele and Ding \(2017\)](#) is nonparametric, but involves a function of risk ratios that has led to some misinterpretations; see e.g. [VanderWeele et al. \(2019\)](#) and [VanderWeele et al. \(2019\)](#).

In this paper, I propose a method that allows researchers to easily relax a broad range of additional assumptions beyond confounding. Intuitively, the approach works as follows. First, a researcher proposes a *initial model*, representing possible data-generating processes (DGPs) that satisfy all assumptions used in their original analysis. Then, they propose a *relaxed model*, a larger set of possible DGPs in which up to proportion θ of observations may violate the initial assumptions. The original analysis effectively supposes that $\theta = 0$. The proposed sensitivity analysis generalizes this by slowly increasing the proportion of assumption violations until $\theta = 1$, which is equivalent to discarding the assumption entirely. After researchers propose these two models and specify a value for θ , the procedure works by calculating *sharp bounds* for an estimand, representing best- and worst-case scenarios while allowing for a θ -level violation of the causal assumptions in question. These bounds are necessary because when key causal assumptions are relaxed, there may be many possible estimates that are equally compatible with the observed data. For example, in an IV study, if defiers—individuals who refuse the treatment only when encouraged to do so—are known to exist in a population satisfying IV assumptions, then the LATE among compliers is no longer *point identifiable*, meaning that a uniquely best estimate no longer exists. However, it is still possible to calculate a *partial identification* region, or bounds, that contain equally good estimates. Moreover, some bounds can be shown to be *sharp*, meaning they represent the narrowest range of guaranteed-

valid possible answers after extracting all information from the available data. By leveraging recent advances in automated partial identification methods with discrete data (Duarte et al., 2023), one can examine how this range of possibilities changes as θ increases, for essentially any assumption that can be represented in a causal graph or written down in terms of the distribution of potential outcomes. This procedure offers immediate benefits in interpretability because the approach focuses solely on the causal assumption—specifically, how often it is believed to be violated—rather than ad-hoc parameters created specifically for a secondary analysis, such as hypothetical risk ratios or regression coefficients involving an unobserved confounder. The approach also allows researchers to investigate further by determining the level of θ at which the bounds lose significance or cross zero, conducting falsification tests to rule out certain values of θ , or by probing the hypothetical best- and worst-case DGPs that correspond to the upper and lower bounds.

More specifically, this paper makes two main contributions. First, the proposed method offers a unified language for addressing three broad types of violations. The first type involves violations of functional-form assumptions, which restrict the possible ways that one variable can influence another. For example, in experiments with noncompliance, the “no defiers” assumption rules out the possibility of two-sided noncompliance, making it a functional-form assumption about the way that encouragement influences treatment uptake. Other examples of functional-form violations that the proposed method can handle involve linearity and no-interaction assumptions. The second type of violation pertains to structural assumptions, which correspond to the non-existence of a causal effect. Here, an important example is the “exclusion restriction” assumption, where a randomized encouragement is supposed to operate solely by shaping treatment uptake, rather than directly influencing the outcome. Finally, the third type involves violations of the “no unobserved confounding” assumptions discussed above. As a running example, I focus on IV analysis because it represents a widely used design that illustrates each of the three assumption types. In other work (Duarte et al., 2023), I showed that essentially all standard causal assumptions can be categorized into one of these three types; the current paper extends that work by demonstrating how partial identification allows any such assumption to be relaxed in a sensitivity analysis. It is important to note

that this is not the first method to analyze the sensitivity of estimands to violations in terms of partial identification bounds. For example, the proposed method is closely related to prior work by [Richardson and Robins \(2010\)](#), which uses principal stratification to probe sensitivity to monotonicity violations, and [Bonvini and Kennedy \(2022\)](#), which considers the proportion of units violating a confounding assumption. In this work, I provide a unified framework that connects and generalize both methods.

The second contribution is this method is automation. I not only provide a general sensitivity analysis method, but also an algorithm to generate sensitivity curves for any causal violation. This algorithm utilizes **autobounds** ([Duarte et al., 2023](#)), an algorithm that generates sharp bounds for any standard estimand in a discrete setting, automatically incorporating any observed data along with any of the assumption types described above. Using the publicly available, researchers can quickly obtain results without the tedious and error-prone step of deriving bounds manually. A potential drawback of **autobounds** is that it produces numeric results based on the observed data, rather than closed-form symbolic solutions that apply to possible future data. These numeric results not always provide the intuition needed for understanding causal problems or conducting statistical inference. To address this limitation, I offer additional symbolic bounds using the method proposed by [Duarte \(2024a\)](#).

This paper proceeds as follows. In Section 2, I introduce a motivating IV application from [Davenport et al. \(2010\)](#), a get-out-the-vote (GOTV) experiment in which a randomized encouragement to participate in election t was used to study the “habit-forming” effects on subsequent participation at $t + 1$. In Section 3, I discuss the theory behind the sensitivity analysis method, explaining how θ -robustness can be evaluated by showing that the direction of a result can be bounded even when proportion θ of observations may involve violations of an assumption. To do so, I first show how standard causal-inference approaches can be reframed as special cases of sharp-bounding problems, which then allow the original identification assumptions to be relaxed. I provide several worked examples that set up and solve a variety of sensitivity problems. In Section 4, I demonstrate the application of this method to derive novel sensitivity analysis results for IV designs and factorial experiments, and I show how the same conceptual framework can also accommodate the confounding problems

that have been a mainstay of existing sensitivity approaches. Finally, in Section 5, I apply the method to a range of real-world problems. Specifically, I replicate an experiment by [Davenport et al. \(2010\)](#), where researchers investigate the role of habit formation in voter turnout—whether participating in an election for any reason increases the likelihood of the same voters participating in subsequent elections. I show that this result is not robust to even a minimal 1% presence of defiers, such as those who would vote but decide not to under canvassing. Next, I apply the method to [Hobolt’s \(2016\)](#) finding that higher education leads to “stay” votes in the Brexit referendum is 0.06-robust to selection-on-observables violations: unobserved common causes of education and votes must be present in 6% of respondents to overturn results. Similarly, findings from [Knox et al. \(2020\)](#) on racially discriminatory police force are 0.06-robust to violations of a monotonic selection assumption: 6% of initial police decisions to stop must be discriminatory against white civilians before conclusions about subsequent use of force cannot be supported. This method offers a widely applicable, easy-to-interpret tool for examining the implications of assumption violations on causal conclusions.

2 A Motivating Example

The literature on habit formation has explored the extent to which GOTV interventions can establish voting as a habitual behavior. Early studies ([Green et al., 2003](#); [Davenport et al., 2010](#); [Coppock and Green, 2016](#)) demonstrated that consistent exposure to GOTV efforts that encourage turnout—such as door-to-door canvassing, phone calls, and mailings—can increase voter turnout, particularly when these interventions are repeated over multiple election cycles. The underlying theory is that voting, like other behaviors, can become habitual when consistently reinforced, especially in the presence of cues such as reminders or social pressure. This perspective draws on behavioral psychology, which suggests that repeated actions in stable contexts can lead to the formation of habits, making voting a more automatic behavior over time.

In methodological terms, the analysis of habit formation often employs instrumental variable strategies, where being encouraged to vote in the first election is the instrument, voting

in this election is the treatment, and voting in the subsequent election is the outcome. In this strategy, researchers aim to identify and estimate the local average treatment effect (LATE [Imbens and Angrist, 1994](#)), which represents the difference in second-election voting rates that would manifest if individuals could be forced to vote in the first one, averaged over compliers who would only vote if encouraged to do so in the first election. To investigate habit formation, [Davenport et al. \(2010\)](#) examines data from several experiments. One key analysis involves individuals who randomly received a social-pressure message encouraging them to vote in an August 2006 primary election—specifically, by demonstrating that a failure to vote could be revealed to their neighbors. Data on voter turnout was collected in that election and the subsequent general election in November 2006.

The most common IV strategy involves a suite of assumptions—most notably, the randomized encouragement, “exclusion restriction,” and “no defiers” assumptions—which will be defined more formally in section 4.1. The first two assumptions are classified as structural, as they correspond to restrictions on the structure of the causal graph. Briefly, the exclusion restriction refers to the GOTV message having no direct influence on general-election voting, while random assignment means that no confounding variable jointly affects the GOTV message and voting in either election. The “no defiers” assumption, on the other hand, is a functional-form assumption that imposes a restriction on the shape of the GOTV message’s influence on primary-outcome voting. Here, “no defiers” implies that individuals may be unaffected by the GOTV message or may comply with its encouragement to vote, but no one would refuse to vote in the primary solely because they were encouraged to do so.

These three assumptions are not always harmless. If any are violated, estimates will generally be incorrect. More importantly, assumption violations could potentially lead analysts to erroneously conclude that an effect exists when in truth it does not; in more severe cases, they may even mislead analysts about the direction of the effect. Consider the exclusion restriction, which requires that every voter who received a GOTV mailer in August must have forgotten its message a mere three months later—or, at least, not let their November turnout be influenced by the knowledge that their neighbors can tell if they violated an important civic norm. In the case of the no-defiers assumption, at least one study has documented a backlash to

canvassing—showing that GOTV efforts can reduce responsiveness and may reduce support for a candidate—suggesting that the existence of defiers is quite plausible (Bailey et al., 2016).

For this reason, it is important to evaluate the sensitivity of the results to violations of key assumptions. How would estimates change if analysts conceded that perhaps 1% of subjects might exhibit defiance, or that up to 10% of subjects might remember a GOTV message in a subsequent election? Similar questions about the robustness of causal inferences can be asked not only about IV analyses, but about other research designs as well. However, there currently exists no general solution for addressing many of the assumptions that underlie the wide variety of designs that are used in applied work. The proposed methodology aims to address this gap.

3 Formalizing the Proposed Sensitivity Analysis Approach

3.1 Notation for the Running IV Example

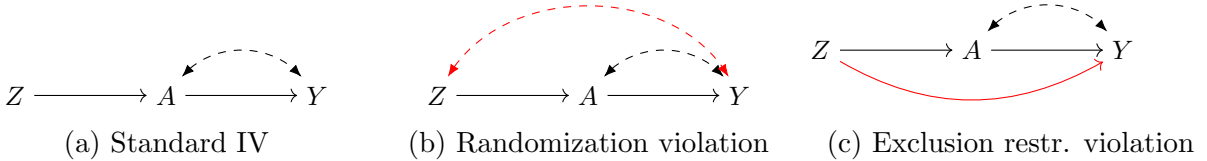
For expository purposes, I begin by introducing notation for the proposed framework in the IV setting. I assume each unit is an independent and identically distributed draw of (X, Z, A, Y) , where $Z \in \{z_0, z_1\}$ is a binary encouragement, $A \in \{a_0, a_1\}$ is a binary treatment, $Y \in \{y_0, y_1\}$ is the binary outcome of interest, and X represents a collection of covariates. While the sensitivity analysis described in this paper is presented for binary variables, the approach holds for categorical variables, and it can be extended to continuous outcomes by discretizing at arbitrary cutoffs as well.¹ Lowercase symbols such as z represent possible realizations of a random variable Z . Additional variables will be introduced as needed for specific cases. Assumptions for identifying the target causal parameters are formulated in potential outcome notation (Rubin, 1974). Here, $Y(a)$ denotes the potential outcome that *would have been observed* if the treatment had been set to a . The standard “stable unit treatment value

¹For instance, sensitivity analyses can be extended to continuous outcomes by bounding the cumulative distribution function $F_{Y(a)}(y) \equiv P(Y(a) \leq y)$ and then integrating over these bounds to obtain $\mathbb{E}[Y(a)]$.

assumption” and positivity assumption are also made throughout (Imbens and Rubin, 2015).

Graphs are also used to encode sets of causal assumptions in our analysis. Specifically, I use directed acyclic graphs (DAGs), which consist of vertices V and directed edges $E \subset V \times V$ such that cycles are not allowed. DAGs represent unconfounded causal models, where each vertex $V_i \in V$ represents a random variable, and an arrow (V_i, V_h) indicates that V_h is a function of V_i (Pearl, 2009). To represent unobserved confounding between two variables V_i and V_h , the DAG structure can be extended with bidirected edges ($E_b \subset V \times V$), resulting in an acyclic directed mixed graphs (ADMG) (Evans, 2016). For example, consider the graph in Figure 1, which represents an instrumental variable scenario with instrument Z . The bidirected edge between A and Y indicates that both treatment and outcome are confounded by unobserved variables. The absence of a bidirected $Z - A$ and $Z - Y$ edges represents the assumption of random assignment, while the absence of a direct arrow from Z to Y represents the exclusion restriction assumption. For LATE identification, the “no defier” assumption is typically required, though it has no direct graphical representation. These three types of violations—existence of a bidirected edge, existence of a direct arrow, and absence of functional restriction—exemplify the three classes of assumption violations discussed in the following section.

Figure 1: IV data-generating process. The ADMG in (a) represents the structural assumptions of a standard IV model. In (a), the random assignment assumption is indicated by the absence of a bidirected edge from encouragement Z to either treatment A or outcome Y . In (b), the red bidirected edge highlights one possible violation of this assumption. Returning to (a), the exclusion restriction assumption is reflected in the fact that the only path from Z to Y passes through A . The direct $Z \rightarrow Y$ edge in (c) represents a violation of this assumption. Finally, “no defiers” is a functional-form assumption that is not depicted in most graphical representations.



3.2 Notation and Theory for General Sensitivity Analyses

In this section, I formalize a widely applicable approach to sensitivity that is based on the proportion of units for which key assumptions may be violated.

The core idea is to report how sharp bounds for any estimand $\mathcal{E}$ vary as the proportion of assumption-violating units increases. One common estimand is the average treatment effect (ATE), $\mathcal{E} = P(Y(a_1) = y_1) - P(Y(a_0) = y_1)$, but others include the local average treatment effect (LATE) among compliers and the risk ratio. This approach allows us to determine the maximum extent of violation that can be tolerated before researchers can no longer confidently sign the causal estimand. One practical challenge in applying the proposed sensitivity method is that there is often ambiguity in defining this estimand (Lundberg et al., 2021).

This sensitivity analysis framework involves several key steps. The first is to formally define the initial causal-inference problem, denoted $\mathcal{S}$, in terms of the pair $(\mathcal{M}, \mathcal{E})$. Here, $\mathcal{M}$ represents a causal model representing the set of possible DGPs over variables $V = \{V_1, \dots, V_I\}$ with known discrete support that satisfy a set of assumed constraints $\mathcal{C} = (C_1, \dots, C_J)$. For example, in the running binary IV example, assumed constraints on the model include a single no-defiers constraint, $C_{\text{ND}} : P(A(z_1) = a_0, A(z_0) = a_1) = 0$; a set of exclusion restriction constraints $C_{\text{ER},a,z} : P(Y(a) \neq Y(a, z)) = 0$ for each a and z ; and a set of unconfoundedness constraints $C_{\text{UC},z,y,y'} : P(z, Y(z_0) = y, Y(z_1) = y') = P(z)P(Y(z_0) = y, Y(z_1) = y')$ for all z , y , and y' . Note that this last set of constraints is merely a reformulation of the assumption that $Z \perp\!\!\!\perp (Y(z_0), Y(z_1))$.

Once the causal problem is defined, the next step involves solving it, which includes two parts. First is the *identification* part, which assesses the extent to which the causal problem can be resolved if researchers knew some information such as the distribution $P(Z, A, Y)$, which would narrow down the model $\mathcal{M}$ to the subspace of possible DGPs that are consistent with that newly collected information. A common approach is to calculate sharp bounds that represent the best- and worst-case scenarios for the estimand $\mathcal{E}$ over the partially identified region (Balke and Pearl, 1994; Duarte et al., 2023). If the lower and upper bounds are equal, the estimand is uniquely determined and the estimand is *point identified* (Shpitser and

Pearl, 2006); otherwise, the estimand is said to be *partially identified*. The second part is the *estimation*, where statistical inference is used to estimate the target estimand as accurately as possible using only finite samples that only provide $\hat{P}(Z, A, Y)$. This process is succinctly summarized by Judea Pearl as “define first, identify second, estimate last” (Van der Laan et al., 2011).

After solving the initial causal problem, a researcher might then question the impact of potential violations of the underlying assumptions. In the causal problem $\mathcal{S}$, suppose there is a subset of questionable assumptions $\tilde{\mathcal{C}} \subset \mathcal{C}$, representing K out of the original J assumed constraints. Our goal is to construct a modified set of constraints that is in one-to-one correspondence with this K -subset of questionable assumptions, but allows for the possibility that they may not be exactly true. To do so, we will begin by observing that many of the initial assumed constraints C_j naturally take on the form of an (in)equality in which the left-hand side is some expression involving the distribution of potential outcomes, such as $P(A(z_1) = a_0, A(z_0) = a_1)$ in the case of the C_{ND} no-defiers assumption, and the right-hand side is zero, meaning that the specified group does not exist.² This can be considered a special case of $\tilde{C}_{\text{ND},\theta} : P(A(z_1) = a_0, A(z_0) = a_1) = \theta$ when $\theta = 0$. To probe this assumption, we will simply allow θ_{ND} to take on larger values that represent a plausible size for the defier group. This process will be repeated for each of the questionable assumptions in $\tilde{\mathcal{C}}$.

I will denote the vector of assumption-violation proportions as $\Theta \in [0, 1]^K$; these correspond to the right-hand-side constants in the modified constraints $\tilde{\mathcal{C}}_\Theta$ that loosen each questionable assumptions by the corresponding value in Θ . It is worth noting that some assumptions are more complex than others. For example, the collection of four exclusion-restriction constraints are handled by combining them into a single pooled constraint;³ confounding-related assumptions can also be formulated in this way with some additional effort. These cases will be considered in more detail below.

In summary, the process described above produces the modified set of assumed constraints

²When the right-hand side initially takes on some value other than zero, this quantity can be subtracted from both sides.

³Specifically, this combined form $P(Y(a_0) \neq Y(a_0, z_0) \cup Y(a_0) \neq Y(a_0, z_1) \cup Y(a_1) \neq Y(a_1, z_0) \cup Y(a_1) \neq Y(a_1, z_1)) = 0$.

$\mathcal{C} - \tilde{\mathcal{C}} \cup \tilde{\mathcal{C}}_\Theta$, where the set difference $\mathcal{C} - \tilde{\mathcal{C}}$ represents the unobjectionable assumptions and the final term replaces the questionable assumption constraints with their modified versions. Next, observe that by relaxing their original assumptions, analysts have created a larger model space $\tilde{M}_\Theta$, which defines a relaxed causal-inference problem $\tilde{\mathcal{S}}_\Theta = (\tilde{M}_\Theta, \mathcal{E})$ for which we can calculate sharp lower and upper *sensitivity bounds*, $L(\tilde{\mathcal{S}}_\Theta)$ and $U(\tilde{\mathcal{S}}_\Theta)$, can be computed. If we solve $\tilde{\mathcal{S}}_\Theta$, then we would know the range of possible answers to the original question $\mathcal{E}$ under a particular proportion of allowed violations for each questionable assumption. Thus, the output of $\tilde{\mathcal{S}}_\Theta$ can be understood as a function of Θ . This defines a perturbation function f .

Definition 1 (Perturbation function). *Let $f : [0, 1]^K \rightarrow [l, u] \subseteq \text{Codomain}(\mathcal{E})$ be the perturbation function $f(\Theta) = [L(\tilde{\mathcal{S}}_\Theta), U(\tilde{\mathcal{S}}_\Theta)]$. This function maps a fixed level of violation Θ for assumptions $\tilde{\mathcal{C}}$ to the interval of admissible estimand values defined by the lower and upper sensitivity bounds $L(\tilde{\mathcal{S}})$ and $U(\tilde{\mathcal{S}})$ if they exist, or the empty interval if there exists no DGP consistent with the given constraints.*

The perturbation function allows researchers to evaluate the impact of a specified level of assumption violation on the conclusions that can be drawn in a causal analysis. For example, consider again the problem of estimating the LATE using the instrumental variable model described above, supposing that the researcher has initially point-identified the estimand under the assumption that the proportion of defiers, $P(A(z_0) = a_1, A(z_1) = a_0) = 0$. For simplicity, suppose that the researcher is confident in other IV assumptions, but wishes to consider a scenario where exactly 20% of units are defiers. In this case, the perturbation level is $\Theta = [\theta_{\text{ND}}]$, where $\theta_{\text{ND}} = 0.2$, and the modified assumption constraint set is $\tilde{\mathcal{C}} = (C_{\text{ND}} : P(A(z_0) = a_1, A(Z_1) = a_0) = 0.2)$. The perturbation function $f(\Theta)$, evaluated at $\Theta = [0.2]$, would then provide the lower and upper bounds for this revised causal problem.

One challenge with this approach is that researchers often lack precise information about the exact proportion of violations in a model. Consequently, the perturbation function might not be directly applicable, except for theoretical evaluations of how solutions change as Θ varies. However, researchers can often make informed guesses about the *maximum* plausible

proportion of violations. In the instrumental variable example, this would involve estimating the maximum possible proportion of defiers. For instance, one might ask how the results would change if the proportion of defiers were allowed to be *at most* 0.2 rather than *exactly* this value. Additionally, researchers can estimate the maximum possible proportion of violations such that the bounds or their confidence interval no longer rule out a zero effect. These considerations open up new avenues for investigating and understanding the robustness of scientific conclusions.

Definition 2 (Sensitivity analysis function). *Let $\Theta^{\max} \in [0, 1]^K$ be a vector of maximum proportions of assumption-violating units for each of the K questionable assumptions in $\tilde{C}$. A sensitivity analysis is a function $g : [0, 1]^J \rightarrow [l, u] \subseteq \text{Codomain}(\mathcal{E})$ mapping $\Theta^{\max}$ to $\bigcup_{\Theta \leq \Theta^{\max}} f(\Theta)$, an interval defined by the minimum lower bound and maximum upper bound over all perturbations.*

The sensitivity analysis function provides the sharp lower and upper bounds based on a specified maximum proportion of violations of a particular causal assumption. It is important to note that for a particular $\Theta^{\max}$, the lower bound may correspond to one perturbation Θ , while the upper bound corresponds to $\Theta' = \Theta$. The curve generated by this function can offer researchers valuable insights into their scientific inquiries, helping them understand how much a specific assumption influences their results. However, obtaining these functions for any given case can be challenging. Below, I describe the automated partial identification system developed by Duarte et al. (2023) and the symbolic solver proposed by Duarte (2024a). I also demonstrate how to represent assumption violations within this system.

3.3 A Brief Primer on Principal Stratification and autobounds

Manually deriving sharp bounds on effects can be analytically complex and time-consuming. To simplify this process, Duarte et al. (2023) offers a user-friendly algorithm that computes sharp bounds automatically based on the parameters of a causal question. I will utilize this algorithm for the sensitivity analyses described above. For a detailed explanation of how this algorithm works, refer to Appendix A.1 or the comprehensive description in Duarte et al.

(2023). Here, I provide an overview of how `autobounds` works.

The algorithm takes four inputs: a causal theory, represented in an ADMG; any additional functional assumption not captured by the ADMG (e.g. “no defiers” or “no interaction”); a causal estimand (e.g. the ATE); and empirical distributions over a set of discrete variables. The algorithm then searches over possible solutions—values of the causal estimand—that are consistent with all these parameters, returning the minimum and maximum of the solutions discovered.

Central to this procedure is the idea of principal stratification (Balke and Pearl, 1997; Frangakis and Rubin, 2002), which involves categorizing units in a study based on their potential responses under different conditions. For instance, in the classic IV framework, units are divided into four principal strata: compliers, defiers, always-takers, and never-takers. These strata reflect how treatment responds to the instrument, but any discrete causal model can be framed in terms of principal strata based on how each relevant variable might behave under various conditions. As the complexity of causal systems increases, the number of principal strata grows rapidly. However, as long as all variables are discrete, the number of strata remains countable and finite, because there are only a limited number of ways discrete variables can interact. Even with continuous unobserved confounders, the number of potential values for the causal estimand remains finite. Additionally, `autobounds` operates in an “anytime” manner, meaning that if the algorithm is stopped before sharp bounds are fully determined, it will still provide valid bounds (Dean and Boddy, 1988).

In formal terms, the method of principal stratification involves considering, for every variable V_i in a graph, all possible functions that map all possible values of all proximal causes (parents in the graph) to possible values of V_i . If V_i can take on c values and its parents have d possible unique combinations of values, then the set of principal strata for V_i , S_{V_i} , has cardinality c^d , where d is the cardinality of the combination of all parents’ levels. For instance, in the instrumental variable scenario with binary variables, A has only one observable parent, Z . Then the strata of A must account for all possible functions from Z to A , which can be compactly written as $S_A = \{a_{00}, a_{01}, a_{10}, a_{11}\}$, representing respectively never-takers, compliers, defiers, and always-takers. In this notation, the elements of S_A are indexed by

binary variables, where the first digit indicates the value of A if Z were 0, and the second digit, if Z were 1. Likewise, $S_Y = \{y_{00}, y_{01}, y_{10}, y_{11}\}$, as Y has only A as an observable parent. On the other hand, Z has no observable parent, so its strata are $S_Z = \{z_0, z_1\}$, respectively representing control and encouraged units. After the variables are stratified, it is still necessary to consider potential unobserved confounders, which can shape the joint distribution of strata $P(S_Z, S_A, S_Y)$. In particular, strata for variables connected by bidirected edges (“c-components,” [Tian, 2002](#)) must be understood as non-separable strata. For example, under standard IV modeling assumptions, randomization of Z means that S_Z is independent of the remaining strata, but confounding of A and Y mean that S_A and S_Y are dependent. Thus, we can only factorize the joint distribution of strata as $P(S_Z, S_A, S_Y) = P(S_Z)P(S_A, S_Y)$. For compactness, and following prior work on principal stratification, I will sometimes refer to non-separable “joint strata” with notation such as $a_{00}y_{01}$, rather than considering separated strata a_{00} and y_{01} . Formal algorithms and a provably complete method for stratification in discrete ADMGs are described in detail in [Appendix A.1](#) and [Duarte et al. \(2023\)](#).

After the original graph is stratified, the goal is to be able to represent every causal statement in that setting, including both data and estimands. For example, if the probability of compliers can be represented exactly by $P(a_{01}y_{00}) + P(a_{01}y_{01}) + P(a_{01}y_{10}) + P(a_{01}y_{11})$, and LATE can be expressed as follows:

Lemma 1. *For a binary instrumental variable model, the LATE estimand $\mathbb{E}[Y(A = 1)|A(z_1) > A(z_0)] - \mathbb{E}[Y(A = 0)|A(z_1) > A(z_0)]$ can be represented as:*

$$\frac{P(a_{01}y_{01}) - P(a_{01}y_{10})}{P(a_{01}y_{00}) + P(a_{01}y_{01}) + P(a_{01}y_{10}) + P(a_{01}y_{11})} \quad (1)$$

For a complete stratification of an instrumental variable, including estimands and data can be found in [Appendix A.2](#). The idea behind this procedure is that we can maximize/minimize the estimand given data and assumptions (constraints) over the causal model strata (parameters).

Employing stratification, `autobounds` efficiently identifies all principal strata suggested by a causal model and its assumptions. It crucially discards strata that cannot exist given

the specified assumptions and observed data, or those that provide redundant information. The program then represents the causal estimand as a polynomial expression based on these principal strata, which can be optimized computationally, subject to the constraints of the causal theory, assumptions, and data. Once in this form, finding sharp bounds becomes a computational problem. The main challenge is achieving solutions within a reasonable timeframe, especially as the number of principal strata grows rapidly with the increasing complexity of the variables involved. Recent advancements by [Duarte \(2024a\)](#) enhance the `autobounds` package by adding a closed-form component, enabling the derivation of analytic bounds for causal estimands using the same four-step approach as its numerical version.

3.4 Analysis of Relaxations for Three Assumption Types

As discussed in previous sections, the proposed method for conducting sensitivity analysis involves a parameter representing the proportion of units that violate each assumption. Here, I describe how the implementation of general idea varies across three types of violations: (a) of functional assumptions, (b) of exclusion restrictions, and (c) of unconfoundedness.

Functional assumptions correspond to constraints on how one variable affects another. If a model assumes that encouragement Z causes treatment A but restricts the form of this causal function, this constitutes a functional constraint. The “no defier” assumption discussed above is a *unit-level* functional-form constraint, as it implies that among all possible responses of the instrument to the treatment, only compliers, never-takers, and always-takers should exist. I will begin by focusing on unit-level functional assumptions here. The method described in the preceding section does not require further explanation for this case, as it applies immediately to unit-level functional assumptions. However, linearity assumptions, such as $\mathbb{E}[Y(W, X)] = \alpha + \beta W + \gamma X$, represent *aggregate-level* functional-form constraints; violations of these will involve a slightly different approach that I will discuss later, as it is similar to the approach used to handle confounding violations.

Next, consider exclusion-restriction assumptions, which rule out the presence of a direct causal influence from a variable X to another variable Y that is not mediated by other

variables. In graphical language, this is represented by the lack of an $X \rightarrow Y$ arrow. For the purposes of sensitivity analysis of violations of these assumptions, I define a causal-inference problem that allows for the existence of this arrow; this means that the strata of Y must now be expanded to include levels representing units for whom the value of Y changes when holding Y 's other parents' values fixed but varying X . As a simple example, suppose that Y has no other parents: in this case, Y has only two strata, $S_Y = \{y_0, y_1\}$. If we allow for an $X \rightarrow Y$ arrow, then Y will have four strata, $S_Y = \{y_{00}, y_{01}, y_{10}, y_{11}\}$. A sensitivity analysis of the X to Y exclusion restriction here will involve the perturbed constraint $P(a_{01}) + P(a_{10}) = \theta$, because these represent the newly added strata that were assumed not to exist under the original constraint. When $\theta = 0$, the new model can be shown to be equivalent to the previous model with no arrow.

Finally, consider unconfoundedness assumptions, i.e., the assumption that two variables X and Y are not confounded by an unobserved variable which affects both variables simultaneously, which is represented by the absence of a bidirected edge between them. In terms of strata, unconfoundedness implies that for each $s_x \in S_X$ and $s_y \in S_Y$, the strata satisfy $P(s_x, s_y) = P(s_x)P(s_y)$, i.e., the strata are independent and therefore have joint probabilities equal to the product of their marginal probabilities. Using the standard strata decomposition, there is no straightforward method to directly set a proportion of violations. For instance, even if $P(s_x, s_y) - P(s_x)P(s_y)$ differs from 0, the magnitude of the difference has no clear interpretation. One possible approach would be to define a sensitivity parameter that restricts the mutual information of the strata, but interpretation of such a parameter would be difficult as well.

To address this issue, we build on an elegant idea first developed by [Bonvini and Kennedy \(2022\)](#). (However, in the next section, I explain how my proposed automated approach allows researchers to avoid the difficult-to-justify simplifications that were used in that work for analytic tractability.) Specifically, we suppose that the observed DGP is a mixture over two graphs: one in which X and Y are indeed unconfounded, with mixture proportion $1 - \theta$, and another in which they are confounded, with proportion θ .⁴ It can be shown that

⁴Mixture models have been used elsewhere in political science to determine the relative explanatory power

this is equivalent to constraining the joint strata distribution to follow $P(s_x, s_y) = (1 - \theta)P^{\text{asm.}}(s_x)P^{\text{asm.}}(s_y) + \theta P^{\text{vio.}}(s_x, s_y)$, where $P^{\text{asm.}}$ is a unspecified hypothetical distribution that satisfies the independence condition implied by the unconfoundedness assumption and $P^{\text{vio.}}$ is an arbitrary joint distribution over the strata. In either case, θ has a straightforward interpretation as the proportion of units drawn from a confounded world, in violation of the assumption. Put another way, if a researcher suspected that an ostensibly randomized experiment was not being carried out according to design, this approach could be used to assess the proportion of times where treatment assignment deviated from plans before the scientific conclusion to no longer be supported. Note that this is not an assumption about the real world, but rather a conceptual tool to formalize the sensitivity problem in terms of the proportion of confounding violations and the resulting sharp bounds.

This method for reasoning about the proportion of units drawn from a confounded DGP was first proposed by [Bonvini and Kennedy \(2022\)](#) for the specific case of treatment-outcome confounding, though that work makes additional assumptions for analytic tractability that are relaxed here. I generalize this method to a graphical representation of the mixture. For any bidirected edge to be analyzed, θ is assumed to be a parameter indicating the mixing proportion of units violating the unconfoundedness assumption, i.e., drawn from a non-independent joint distribution over principal strata of variables touched by the bidirected edge. To see the value of this generalization, consider the linearity assumption described earlier, which can be reformulated as requiring the effect of X on Y to be identical regardless of whether W is set to w_0 or w_1 : $\mathbb{E}[Y(x_1, w_0)] - \mathbb{E}[Y(x_0, w_0)] = \mathbb{E}[Y(x_1, w_1)] - \mathbb{E}[Y(x_0, w_1)]$. Rearranging terms reveals that this also restricts the effects of W on Y when fixing X . It can be shown that this is equivalent to requiring that the assumption-satisfying distribution over Y -strata must follow the constraint

$$2P^{\text{asm.}}(y_{1001}) + P^{\text{asm.}}(y_{0001}) + P^{\text{asm.}}(y_{1000}) + P^{\text{asm.}}(y_{1011}) + P^{\text{asm.}}(y_{1101}) \\ - 2P^{\text{asm.}}(y_{0110}) - P^{\text{asm.}}(y_{0010}) - P^{\text{asm.}}(y_{1110}) - P^{\text{asm.}}(y_{0100}) - P^{\text{asm.}}(y_{0111}) = 0,$$

of competing theories ([Imai and Tingley, 2012](#)). The method used here differs in that the unconfounded model is a subset of the confounded one, meaning symmetric tests cannot be constructed to distinguish between them.

in which each term represents the size of a *unit-level interactive strata*—discussed in more depth in Section 4.3—in which a unit’s outcome depends on the joint values of x and w , rather than one input alone. Effectively, these unit-level interactive strata must counterbalance each other, so that the aggregate-level assumption will hold true. To relax this assumption, again suppose that $P^{\text{asm.}}$ has mixture weight $1 - \theta$, and there exists another unrestricted distribution $P^{\text{vio.}}$ with mixture weight θ . In the linearity setting, intuitively, this is equivalent to allowing an imbalance of up to θ between the opposing groups of interactive strata. This example illustrates how the violation-mixture formulation is rich enough to accommodate a variety of complex assumptions.

Thus, in general, a sensitivity analysis procedure can be conducted to show how sharp bounds on an estimand of interest vary according to violations of each assumption. As demonstrated, there are three types of violations related to three different types of assumptions: (a) functional, i.e., constraints over the functions from causes to effects; (b) exclusion restriction, i.e., assumptions that some variables do not cause directly others; and (c) unconfoundedness, i.e., assumptions that some variables are not influenced by unobserved confounders. These distinctions suggest a new graphical representation to indicate each assumption one would like to test for violations. To represent the sensitivity constraints, we generate sensitivity nodes, which point to the assumptions they are intended to test. Examples of these three types are presented in Table 1. Sensitivity nodes imply d-separation and, consequently, independence. Any node pointing to a direct or bidirected edge will be dependent on both connected variables, while the functional violation node will be dependent only on the variable it is pointing to. Although our examples focus on the sensitivity of only one assumption at a time, this method is sufficiently general to accommodate multiple assumptions simultaneously.

3.5 Solving the sensitivity analysis problem

To carry out sensitivity analysis for violations, I employ two different methods. The first method generates numerical bounds and is complete. It relies on the `autobounds` algorithm and software package introduced in Duarte et al. (2023). In this method, the derivation of per-

Table 1: **Illustration of assumption types in the IV model.** Three types of assumptions and corresponding violations are depicted; the proposed method can perform sensitivity analysis on each of these types. The functional assumption violation corresponds to constraints from causes to particular variables. For instance, in the instrumental variable example, for LATE identification, a monotonicity or “no defier” assumption is often made ($A(z_1) \geq A(z_0)$, represented with a $+$ sign in the original model graph). If one desires to perform sensitivity analysis on this assumption, it can be relaxed using a parameter $\theta_{\text{ND}} \equiv P[A(z_1) < A(z_0)]$, representing the maximum size of the defier group. Sensitivity analysis on an exclusion restriction assumption, which states that $Y(a, z) = Y(a)$, can be carried out by first assuming the model with the arrow and introducing a parameter $\theta_{\text{ER}} \equiv P[Y(a, z) \neq Y(a)]$, representing the proportion of units for which a direct effect exists. Finally, the third type of violation concerns assumptions of unconfoundedness, i.e., there is no unobserved confounding between variables. θ_{UC} will indicate then the proportion of non-experimental units drawn from an alternative DGP in which this confounding exists.

Assumption	Original Model	Violation
Functional	$Z \xrightarrow{+} A \xrightarrow{\quad} Y$	$Z \xrightarrow{\quad} A \xrightarrow{\quad} Y$ 
Exclusion Restriction	$Z \xrightarrow{\quad} A \xrightarrow{\quad} Y$	$Z \xrightarrow{\quad} A \xrightarrow{\quad} Y$ 
Unconfoundedness	$Z \xrightarrow{\quad} A \xrightarrow{\quad} Y$	$Z \xrightarrow{\quad} A \xrightarrow{\quad} Y$ 

turbation and sensitivity functions is not required, as recent versions of `autobounds` are now sufficiently flexible to directly solve the $g(\Theta^{\max})$ sensitivity problem given in Definition 2. To numerically derive the sensitivity curve, one can simply the exercise for varying values of $\Theta^{\max}$. For each $\Theta^{\max}$, we obtain the sensitivity bounds $\min_{\Theta \leq \Theta^{\max}} L(\tilde{\mathcal{S}}_{\Theta})$ and $\max_{\Theta \leq \Theta^{\max}} U(\tilde{\mathcal{S}}_{\Theta})$; our goal is to find the *critical surface* of transition points, or minimal values of $\Theta^{\max}$ at which $0 \in \left[\min_{\Theta \leq \Theta^{\max}} L(\tilde{\mathcal{S}}_{\Theta}), \max_{\Theta \leq \Theta^{\max}} U(\tilde{\mathcal{S}}_{\Theta}) \right]$. At this point, our estimates suggest we can no longer determine if the effect is positive or negative; a similar exercise can be conducted for the confidence region of the sensitivity bounds. If $0 \notin \left[\min_{\Theta \leq \Theta^{\max}} L(\tilde{\mathcal{S}}_{\Theta}), \max_{\Theta \leq \Theta^{\max}} U(\tilde{\mathcal{S}}_{\Theta}) \right]$, then then we say an effect is $\Theta^{\max}$ -robust.

Although numerical bounds can be useful, there are known challenges of using them in partial identification. Firstly, there are challenges related to statistical inference, which are discussed in section 3.6. Furthermore, understanding the perturbation and sensitivity functions can yield more intuition about the causal problem one is trying to solve. For those reasons, it is important to obtain closed-form solutions when possible. To obtain those closed-form solutions, one can use techniques such as Balke and Pearl (1994), Sachs et al. (2023), and Duarte (2024a) to solve causal programs symbolically. In this paper, I analytically derive results for cases of classic confounding and IV, while conducting numerical sensitivity analysis for more complex scenarios.

3.6 Statistical Inference

So far, I have considered the case where researchers are supposed to observe population distributions of the relevant variables. In reality, the challenge is more complicated due to finite sampling issues. Inference for partial identification regions has been studied for more than two decades in econometrics and statistics (Manski and Pepper, 1998; Andrews and Han, 2009; Manski and Pepper, 2009; Bugni, 2010; Canay, 2010; Chernozhukov et al., 2013; Levis et al., 2024; Ji et al., 2023). The literature suggests several issues with methods such as the simple bootstrap, which I briefly review here.

First, because bounds usually involve maxima or minima over a set of functionals of

probability distributions, it is known that sample analogs of those parameters exhibit finite-sample bias, which increases with the number of formulas included (Manski and Pepper, 2009). This type of bias is less of a concern with large sample sizes, as the estimators considered here are extrema of a fixed number of simple formula, each of which can be well-approximated by normal distributions, and those extrema can be shown to be consistent. Alternatively, one can estimate the maximum or minimum of m -varied normal distributions (Cai et al., 2007) or debiasing the sample analog by subtracting a correction of the form $\sigma \cdot \sqrt{2 \log m}$, where m is the number of functionals.

Secondly, it is known that if multiple functionals involved in a minimum or maximum happen to take on the same value for a given population distribution, the bounds will change non-differentiably as the population distribution shifts. In turn, this creates issues when deriving the asymptotic distribution. For instance, Andrews and Han (2009) shows that if we intend to estimate $\min \mu_1, \mu_2$ for expected values of two different independent normal distributions, the asymptotic distribution of the sample analog will differ depending on whether $\mu_1 = \mu_2$ or $\mu_1 \neq \mu_2$. While there exist approaches that have been shown to yield valid confidence regions, such as that proposed by Duarte et al. (2023), this work has shown they are highly conservative.

While these concerns are theoretically sound, it is also known that the set of DGPs with this non-differentiable boundary issue generally have measure zero in the model space. As a result, recent work such as Levis et al. (2024) has sometimes included a *margin condition*, which essentially states that a technique only possesses certain guarantees if the true DGP does not lie on one of the boundaries. Under this condition, bootstrap methods lead to asymptotically valid intervals as well. Here, I follow this approach as well for simplicity, but it is important to note that if the condition is not satisfied, then results may be anticonservative. More generally, however, I recommend that researchers consider statistical inference problems on a case-by-case basis and make choices according to assumptions that can be properly justified.

4 Worked Examples of Sensitivity Results for Common Designs

To demonstrate the applications of the method, I derive results for standard cases in causal inference. The first case involves instrumental variable analysis. The second addresses violations of the unconfoundedness assumption. Finally, the last case deals with a more complex scenario related to factorial experiments.

4.1 Instrumental Variables

Consider an instrumental variable setting with a binary variable, as shown in Figure 1. For Z to be a valid instrument, it must be effectively randomized, possibly after adjusting for measured covariates X . Additionally, it should only affect the outcome Y through its impact on A . These conditions can be formally stated as:

(A1) Exogeneity: For all $z \in z_0, z_1$ and $a \in a_0, a_1$, $Z \perp\!\!\!\perp (A(z), Y(z)) \mid X$. This is often called the “unconfoundedness” assumption.

(A2) Exclusion restriction: $Y(z, a) = Y(z', a)$ for all $z, z', a \in 0, 1$, meaning that Z affects Y only through its influence on A . In other words, $Y(z, a) \equiv Y(a)$ for all $z, a \in 0, 1$.

The exogeneity assumption is generally assumed to be valid due to randomization in treatment settings, although sensitivity analyses for this assumption in observational studies have been suggested. The exclusion restriction implies that the treatment assignment affects the outcome only through its impact on the treatment variable. Rather than the ATE, it is extremely common for IV studies to target the *local* average treatment effect (LATE) among “compliers” for whom $A(z_1) > A(z_0)$.

$$\text{LATE} \equiv \mathbb{E}[Y(a_1) - Y(a_0) \mid A(z_1) > A(z_0)]. \quad (2)$$

[Imbens and Angrist \(1994\)](#) demonstrated that the traditional two-stage least squares (2SLS)

estimator used in instrumental variable studies can identify the causal effect for specific subgroups. When both the instrument and the treatment are binary, observations can be categorized into four distinct groups, or principal strata: *compliers*, who have $A(1) > A(0)$; *always-takers*, with $A(1) = A(0) = a_1$; *never-takers*, with $A(1) = A(0) = a_0$; and *defiers*, who have $A(1) < A(0)$ (Angrist et al., 1996b; Frangakis and Rubin, 2002). To identify the Local Average Treatment Effect (LATE), researchers commonly add the following assumption:

$$\textbf{(A3) No defiers: } P(A(z_1) < A(z_0)) = 0.$$

The Local Average Treatment Effect (LATE) reflects the average impact of a treatment on the outcome specifically for those units influenced positively by the instrument. Importantly, the LATE can differ significantly from the Average Treatment Effect (ATE) if the effects in other strata differ from those experienced by compliers.

I introduce new sensitivity analyses that help researchers evaluate the robustness of their findings when these assumptions are potentially violated to varying extents. Researchers have previously explored the sensitivity of instrumental variable (IV) results to potential confounding of the instrument (Small and Rosenbaum, 2008). More recently, scholars have started to relax other IV assumptions (Cinelli and Hazlett, 2022; Takatsu et al., 2023), often by imposing linearity or homogeneity constraints on the outcome model. In this work, we propose a set of straightforward sensitivity analyses that directly address the proportion of units that deviate from assumptions of no defiers (also called monotonicity), exclusion, and random assignment.

In essence, my method involves determining precise bounds for the Local Average Treatment Effect (LATE) based on the proportion of units that are defiers, that violate the exclusion restriction, or that breach random assignment. This approach allows me to evaluate the maximum extent of violations that can be tolerated before the causal estimate becomes unidentifiable. More comprehensive results related to instrumental variables are discussed in a separate paper (Cooper et al., 2024).

Assume initially that we have a binary instrument. To assess the sensitivity of our model to the monotonicity assumption, let $\theta_{\text{ND}} \equiv P(A(z_1) < A(z_0))$ represent the proportion of units

that do not adhere to the monotonicity assumption under a given probability distribution. A similar result for this scenario was first derived by [Richardson and Robins \(2010\)](#). Using the type 1 violation approach, it can be demonstrated that:

Theorem 1 (Sensitivity bounds on the LATE for monotonicity violations). *Sharp bounds for the LATE are given by $[l/P(A(z_1) > A(z_0)), u/P(A(z_1) > A(z_0))]$:*

$$l = \max \left\{ \begin{array}{l} -P(y_1, a_0|z_0), \\ P(y_1|z_1) - P(y_1|z_0) - P(y_1, a_0|z_1) \\ P(y_1, a_1|z_1) - P(a_1|z_0) - P(y_1, a_0|z_0) + \theta_{ND}, \\ P(a_1|z_0) - P(a_0|z_0) - \theta_{ND}, \\ P(y_1, a_0|z_1) - P(y_1, a_0|z_0) - \theta_{ND}, \\ P(y_0, a_1|z_0) - P(y_0, a_1|z_1) - \theta_{ND}, \\ P(y_1|z_1) - P(y_1|z_0) - \theta_{ND} \\ -P(y_0, a_1|z_1), \\ P(y_0, a_0|z_0) - P(y_0, a_0|z_1) - P(y_0, a_1|z_1), \end{array} \right.$$

$$u = \min \left\{ \begin{array}{l} P(y_1|z_1) + P(y_0, a_0|z_1) - P(y_1, a_0|z_0) - \theta_{ND}, \\ P(y_1|z_1) - P(y_1|z_0) + P(y_1, a_1|z_0), \\ P(y_0|z_0) - P(y_0, a_1|z_1), \\ P(y_1|z_1) - P(y_1|z_0) + \theta_{ND}, \\ P(y_0, a_0|z_0), \\ P(a_1|z_1) - P(a_1|z_0) + \theta_{ND}, \\ P(y_0, a_0|z_0) - P(y_0, a_0|z_1) + \theta_{ND}, \\ P(y_1, a_1|z_1), \\ P(y_1, a_1|z_1) - P(y_1, a_1|z_0) + \theta_{ND}, \end{array} \right.$$

and

$$P(A(z_1) > A(z_0)) = P(a_1|z_1) - P(a_1|z_0) + \theta_{ND}$$

Proof. Closed-form results can be obtained by solving the symbolic linear program defined in [Appendix B.2](#) □

Now consider the model described in [Figure 1b](#). Following a similar logic to the previous test, we can calculate the same bounds for this model with a parameter θ_{ER} denoting the proportion of units violating the exclusion restriction. For example, when all observed, factual

variables are binary, θ_{ER} is given by

$$\begin{aligned}\theta_{ER} \equiv & 1 - \Pr(Y(z_0, a_0) = y_0, Y(z_0, a_1) = y_0, Y(z_1, a_0) = y_0, Y(z_1, a_1) = y_0) \\ & - \Pr(Y(z_0, a_0) = y_0, Y(z_0, a_1) = y_1, Y(z_1, a_0) = y_0, Y(z_1, a_1) = y_1) \\ & - \Pr(Y(z_0, a_0) = y_1, Y(z_0, a_1) = y_0, Y(z_1, a_0) = y_1, Y(z_1, a_1) = y_0) \\ & - \Pr(Y(z_0, a_0) = y_1, Y(z_0, a_1) = y_1, Y(z_1, a_0) = y_1, Y(z_1, a_1) = y_1) \quad (3)\end{aligned}$$

The summation on the right-hand side is the proportion of units for whom Z has no direct effect on Y . To see this, note that within the probability statement, $Y(z, a) = Y(z', a)$ for each z, z' and a . Therefore, $\theta_{ER} = 0$ is a restatement of Assumption (A2). We finally remark that the formula in (3) naturally extends to many valued instruments and treatments.

Using the type 2 violations strategy, I present a sensitivity procedure to probe the robustness of inferences to partial violations of the exclusion restriction (A2), when “defiers” are assumed to not exist.

Theorem 2 (Sensitivity bounds on the LATE for exclusion violations). *If assumptions (A1) and (A3) hold, then the LATE must satisfy*

$$\frac{1}{P(A(z_1) > A(z_0))} \sum_{i \in \{0,1\}} l_z \cdot P(z) \leq LATE \leq \frac{1}{P(A(z_1) > A(z_0))} \sum_{i \in \{0,1\}} (1 - l_z) \cdot P(z),$$

$$l_z = \max \begin{cases} P(a_1|z_0) - P(a_1|z_1), \\ -P(a_z y_{1-z}|Z_z), \\ P(y_1|z_1) - P(y_1|z_0) - \theta_{ER}, \\ P(a_1, y_0|z_0) - P(a_0, y_0|z_0) + P(a_0, y_0|z_1) - P(a_1, y_0|z_1) - \theta_{ER}, \\ P(a_0, y_1|z_0) - P(a_1, y_1|z_0) + P(a_0, y_1|z_1) - P(a_1, y_1|z_1) - \theta_{ER} \end{cases}$$

and

$$u_z = \min \begin{cases} P(A = z, Y = z|Z_z), \\ P(a_1|z_1) - P(a_1|z_0), \\ P(a_0, y_1|z_0) - P(a_1, y_1|z_0) + P(a_1, y_1|z_1) - P(a_0, y_1|z_1) + \theta_{ER}, \\ P(a_1, y_0|z_0) - P(a_0, y_0|z_0) + P(a_1, y_0|z_1) - P(a_0, y_0|z_1) + \theta_{ER}, \\ P(y_1|z_1) - P(y_1|z_0) + \theta_{ER}, \end{cases}$$

with

$$P(A(z_1) > A(z_0)) = P(a_1|z_1) - P(a_1|z_0)$$

Proof. Closed-form results can be obtained by solving the symbolic linear program defined in Appendix [B.2](#) □

I emphasize that the sensitivity bounds for each estimand presented above are functions of the proportion of units violating either monotonicity, encapsulated by θ_{ND} , or the exclusion restriction, θ_{ER} . However, researchers are often not interested in the sensitivity bounds for an exact proportion of units which violate an assumption, since it is challenging to argue that as a knowable quantity. Rather, it is more intuitive to study the sensitivity bounds above by varying how much uncertainty the researcher possesses about the strength of the violation of the assumption. For example, rather than computing bounds for $\theta_{\text{ND}} = \theta_{\text{ND}}^* \in [0, 1]$, we evaluate bounds on the estimand when $\theta_{\text{ND}} \leq \theta_{\text{ND}}^*$. Fortunately, using the functions derived above, one can easily extend the sensitivity analysis to this approach by using a grid search. In particular, if we divide the space of 0 to θ_{ND} in many intervals, then we can obtain the value of the sensitivity function for each, and then obtain the maximum or minimum of them.

4.2 Classic Confounding

Consider the case where there is a binary treatment and a binary outcome, both of which are confounded by unobserved variables. This scenario is common in observational studies, where researchers often hope to rely on the assumption that all confounders are observed (selection on observables). If the observed confounders are not sufficient to make the treatment independent of the potential outcome, then the effects are not point identifiable, and any estimates will be subject to omitted variable bias. Similar to other cases discussed in this paper, researchers might want to understand how their results could change if some type of violation is recognized.

Despite its simplicity, this problem has posed challenges for decades of scientists. [Cornfield et al. \(1959\)](#), analyzing the observational association between tobacco use and cancer, challenged the hypothesis that a particular hormone was simultaneously increasing both smoking and cancer risk, arguing that for this to be true, hormone levels would have to be at least nine times higher—an implausibly large difference—in smokers than in non-smokers. While Cornfield focused on a specific confounder (hormone), this idea was recently extended by [Van-](#)

derWeele and Ding (2017) to cover cases where the confounder is nonparametric. Despite the straightforwardness of their method, the e-value, there are questions about its interpretability and sharpness. For instance, while it’s relatively simple to conceptualize the association between the treatment and unobservable confounders when these are binary, understanding this with nonparametric confounders is more challenging. Moreover, concerns have been raised that VanderWeele and Ding (2017)’s methods may not fully utilize the observed data, as they rely solely on risk ratios rather than extracting constraints from the entire dataset (MacLehose et al., 2021; Sjölander, 2020; Sjölander and Greenland, 2022). Recent work by Sjölander (2024) has addressed this by deriving sharp results. There remains a trade-off between making parametric assumptions, ensuring interpretability, and achieving sharp results, which is not easily resolved. This explains the variety of methods for assessing sensitivity to simple confounding in the literature, such as those by Rosenbaum and Rubin (1983); Cinelli and Hazlett (2020, 2022); Bonvini and Kennedy (2022), among others.

Building on the theoretical framework of this paper, I propose deriving sensitivity results for a possible proportion of non-experimental units. Bonvini and Kennedy (2022) employed a mixture model for confounding, where the population units are categorized into two types: those that violate unconfoundedness and those that do not. In other words, the confounding model is a causal scenario involving only the treatment and outcome, where potential violations are of type 3, as shown in Table 1. Besides the mixture model, they introduced two additional, highly restrictive assumptions; they explain that these restrictions are simply to make the proposed method computationally tractable, and no substantive justification is offered. The assumptions are (a) that conditional on the distribution of observed covariates X , the distributions of treatment and potential outcomes $Y(a)$ are identical in both the no-confounding DGP and the confounded DGP; or (b) that conditional on the distribution of observed covariates X and on the treatment, the distribution of potential outcomes $Y(a)$ is identical in both the no-confounding DGP and the confounded DGP. As these assumptions are implausibly restrictive and unrelated to the researcher’s goal of addressing the confounding of substantive concern, I derive new closed-form bounds for cases where these assumptions are not guaranteed to hold.

Theorem 3 (Sensitivity bounds under classic confounding). *A valid sensitivity function for $P(Y(A = a) = y)$ in the classic confounding scenario is*

$$\min\{\theta_{UC} \cdot \Delta_{lb}^1 + (1 - \theta_{UC}) \cdot \Delta_{lb}^2, \Delta_{lb}^3\} \leq P(Y(A = a) = y) \leq \max\{\theta_{UC} \cdot \Delta_{ub}^1 + (1 - \theta_{UC}) \cdot \Delta_{ub}^2, \Delta_{ub}^3\} \quad (4)$$

where

$$\Delta_{lb}^1 = \max\left\{1, \frac{P(Y = y, A = a)}{\theta_{UC}}\right\},$$

$$\Delta_{ub}^1 = \min\left\{0, 1 - \frac{P(Y = 1 - y, A = a)}{\theta_{UC}}\right\},$$

$$\Delta_{lb}^2 = \begin{cases} \frac{P(A=a, Y=y) - \theta_{UC}}{P(A=a) - \theta_{UC}}, & \text{if } \frac{P(A=a, Y=y) - \theta_{UC}}{P(A=a) - \theta_{UC}} \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$\Delta_{ub}^2 = \begin{cases} \frac{P(A=a, Y=y)}{P(A=a) - \theta_{UC}}, & \text{if } \frac{P(A=a, Y=y)}{P(A=a) - \theta_{UC}} \in [0, 1] \\ 1, & \text{otherwise} \end{cases}$$

$$\Delta_{lb}^3 = \max\{(1 - \theta_{UC}) \cdot P(Y = y|A = a), P(Y = y, A = a)\},$$

and

$$\Delta_{ub}^3 = \min\{\theta_{UC} + (1 - \theta_{UC}) \cdot P(Y = y|A = a), 1 - P(Y = 1 - y, A = a)\}$$

A proof for this theorem can be found in Appendix B.1. With this formulation, one can construct sensitivity bounds, which will be a complete sensitivity procedure as the function is monotonic. In subsection 5.2, I show an application to the analysis of the impact of education on voting for Brexit (Hobolt, 2016).

4.3 Factorial Experiments

As I demonstrated, the sensitivity analysis procedure proposed in this paper is not limited to instrumental variables or simple confounding but can accommodate any causal model. Consider a factorial experiment with two treatments, A and B , where the goal is to evaluate the effect of each treatment on an outcome Y . For example, A and B might represent gender

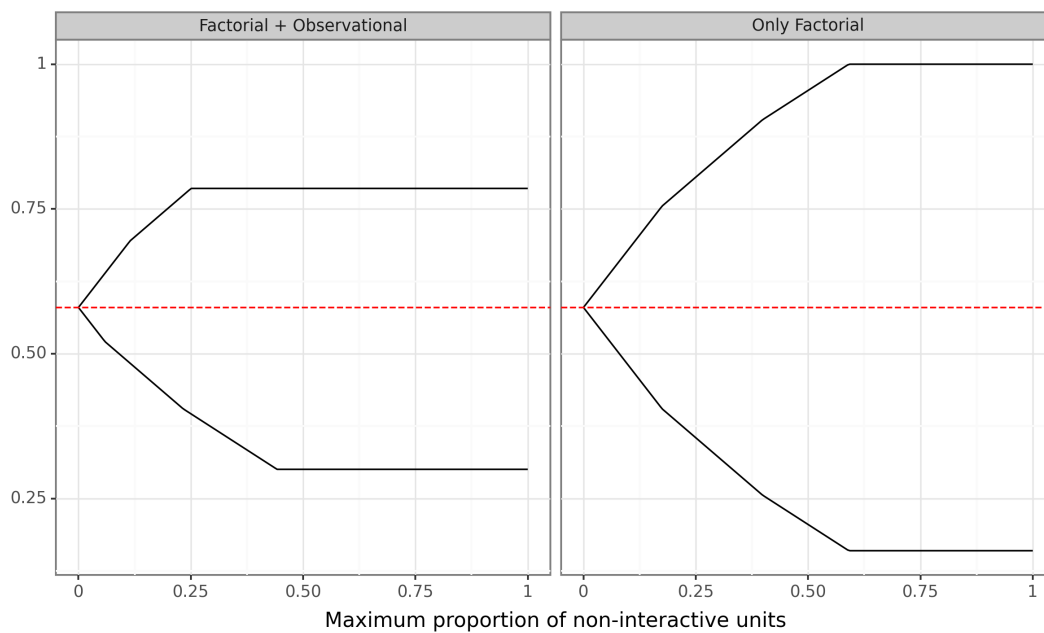
and race, respectively, and the objective is to examine their effects on voting.

If the estimand of interest is the single-treatment average treatment effect (ATE) of A on Y , and both A and B are confounded with Y outside the experimental design, it is known that point-identifiability cannot be achieved without major assumptions (Duarte, 2024b). The intuition behind this result is that the factorial experiment manipulates two treatments simultaneously, but the estimand concerns a scenario where there is an intervention on A , while B remains confounded with Y . Consequently, even when combining factorial data with observational data, point-identification of the single-treatment ATE is not possible. The “no interaction” assumption with respect to B and A on Y is sufficient for point-identifiability of the ATE. However, what if there is an interaction between A and B for certain units of the population? This can be evaluated through sensitivity analysis.

(A4) No interaction: $y_i = h(a_i, u_i) + l(b_i, u_i)$ with unobserved u_i for arbitrary fixed functions h, l is a real-world model.

In contrast to the previous case involving instrumental variables, I only derive numerical results here. In this scenario, the strata associated with interaction are restricted to a maximum of θ_{NI} , while the rest are considered freely. Using the notation y_{rstu} for the strata $Y_{a_0, b_0} = r$, $Y_{a_0, b_1} = s$, $Y_{a_1, b_0} = t$, $Y_{a_1, b_1} = u$, there are six non-interactive strata: y_{0000} , y_{0011} , y_{0101} , y_{1010} , y_{1100} , y_{1111} , with all others being interactive. Using `autobounds` to handle the problem’s intricacies, sensitivity curves can be generated for both cases: (a) when factorial and observational data are combined and (b) when only factorial data is used. Curves using simulated data are shown in Figure 2. In both cases, bounds continuously widen in a piecewise linear fashion, then stop widening after around 50% of non-interactive units. Practically, this indicates that non-interaction assumptions are crucial for point-identification, and the positive effect allows for significant relaxations of the assumption. This example demonstrates that the method proposed here can be applied to more complex models and can lead to the development of new theories.

Figure 2: Sensitivity analysis of the ATE, constrained by the proportion of non-interactive units, is presented. The details of the model are introduced in [Duarte \(2024b\)](#). The dashed line represents the actual ATE of the model (0.58), while the black lines depict strict bounds corresponding to varying constraints on the maximum proportion of non-interactive units (x-axis). A maximum constraint of 0 implies no interaction, resulting in exact point identification of the ATE.



5 Empirical Results

5.1 Instrumental Variables: Reanalysis of [Davenport et al. \(2010\)](#)

I now replicate and extend a prominent get-out-the-vote (GOTV) study, [Gerber et al. \(2008\)](#), which tested whether various forms of social pressure caused people to turn out in a 2006 primary election in Michigan. A more detailed version of this replication can be found in [Cooper et al. \(2024\)](#). Specifically, voters were randomly assigned to be informed that their turnout activity would be monitored by researchers and that their neighbors would be informed as to whether they voted. Since its publication, this study has been re-analyzed in the instrumental variables framework to test whether voting is “habit forming,” ([Gerber et al., 2003](#); [Davenport et al., 2010](#); [Coppock and Green, 2016](#)). In this revised setup, the initial social pressure intervention is treated as a binary instrument (encouragement, Z) to vote in the 2006 primary election (which is conceptualized as the treatment, A), but the outcome of interest, Y , is whether people vote in a subsequent election. In other words, this re-analysis tests whether the experience of voting in election t increases the probability of voting in election $t + 1$.

There are reasons to doubt that monotonicity holds in this setting: given the intrusive nature of the intervention, it is quite possible that social pressure caused some voters not to turn out in election t who otherwise would have. In addition, the exclusion restriction implies that habit formation is the only mechanism through which turnout would increase in election $t + 1$: that is, social pressure increases turnout only insofar as it affects turnout in election t , and there is no direct channel through which social pressure affects downstream turnout. However, as [Davenport et al. \(2010\)](#) notes, another mechanism, “social learning,” whereby the content of the encouragement intervention causes people to internalize the civic norm of voting, is also possible ([Bandura and Walters, 1977](#)).

If both assumptions are correct, then the LATE is 0.124, with 95% CI of [0.08, 0.17]. This raises the question of just how large a violation would have to be to neutralize these results, i.e. to prevent us from signing the the LATE.

To investigate, I conduct the analysis from Section 3.2 by computing bounds on the LATE

under various conditions, beginning with monotonicity. Under the classic IV model, monotonicity assumes that $\theta_{\text{ND}} = 0$, i.e. that no units respond negatively to the instrument (i.e. no defiers). It is straightforward to relax this assumption: we can simply input a restriction that sets θ_0 less than or equal to an arbitrary value between zero and unity and recompute bounds on the quantity of interest, assuming that all other IV assumptions hold.

Figure 3 shows the results of this sensitivity analysis for the LATE. In the graph, I illustrate the varying estimates with bootstrap-estimated 95% confidence region over the lower and upper bounds. In particular, I collected bootstrap resamples of the dataset and calculated the 2.5% quantile and the median for the lower bounds, and the median and the 97.5% for the upper bound. Figure 3 shows that it is still possible to sign the LATE as positive so long as no more than 1% of individuals are defiers. Beyond that percentage, however, sharp bounds on the LATE contain zero, and one cannot rule out that the effect of voting in election t among compliers is negative.

I next turn to the most often debated assumption of the IV design, the exclusion restriction, and show how to compute bounds on causal estimands as I gradually relax this constraint. In this analysis, I assume a DAG of the form 1(b), which shows a direct effect between the instrument and the outcome. The proportion of units violating exclusion restriction by θ_{ER} . Figure 4 shows the results of this exercise, displaying bounds on the LATE. Similar to Figure 3, I find that the LATE cannot be signed if more than 1% of units violate the exclusion restriction if the point estimate is considered and 0.65%, if the confidence region is considered. Interestingly, both results are similar despite the different forms of each sensitivity function.

The assumption of random assignment, despite being passible to be checked with sensitivity analysis, can be taken for granted by design, as I replicate results from an experiment where the instrument is randomized. For that reason, there is no justification to believe there is any unit that is not experimental in the data, which precludes us from running other tests.

Figure 3: Sensitivity analysis of LATE with respect to violations of monotonicity. The proportion of units violating monotonicity, i.e. the proportion of defiers (θ_{ND}) is represented on the x-axis. The bounds on LATE change from a point-identifiable estimate when θ_{ND} is 0 to large bounds as the assumed maximum proportion of defiers increases. All bounds are estimated using bootstrap with 1000 resamples. Solid lines over the graph indicate the point estimate of bounds, while the dashed ones represent the 2.5% and the 97.5% quantiles for respectively the lower and the upper bounds of the resamples. The solid vertical line in the graph indicates the value of $\theta_{\text{ND}} = 0.01$ for which the estimated lower bound crosses 0 for the point-estimate, while the dashed vertical line indicates the point where the confidence region crosses 0 at $\theta_{\text{ND}} = 0.0065$. In other words, if the number of defiers is assumed to be above 1% for the point estimate and 0.65% for the confidence region, that means the estimated effect cannot be detected to be positive.

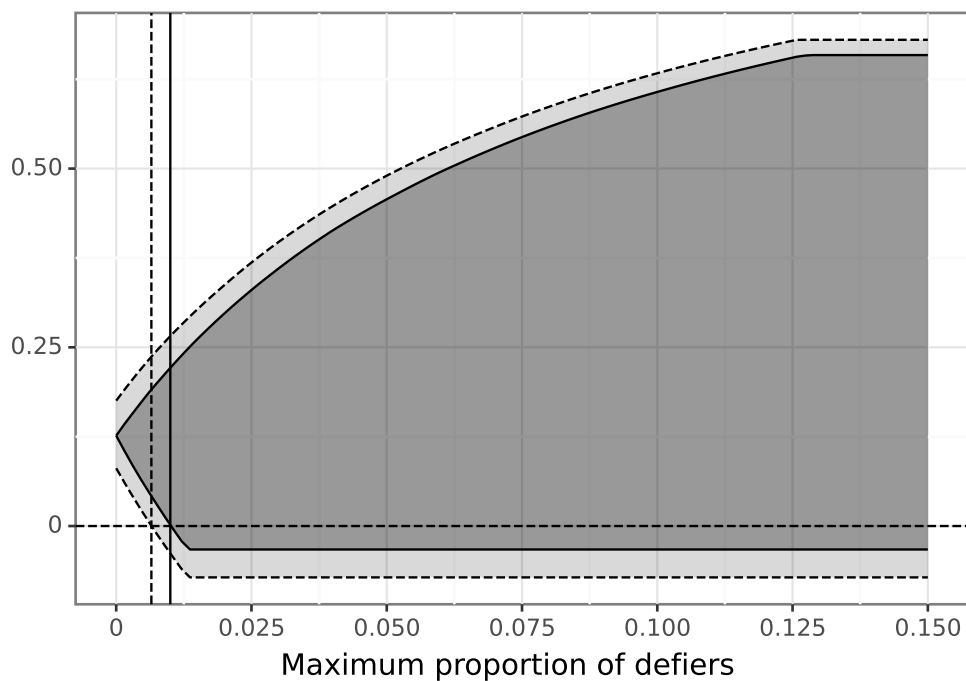
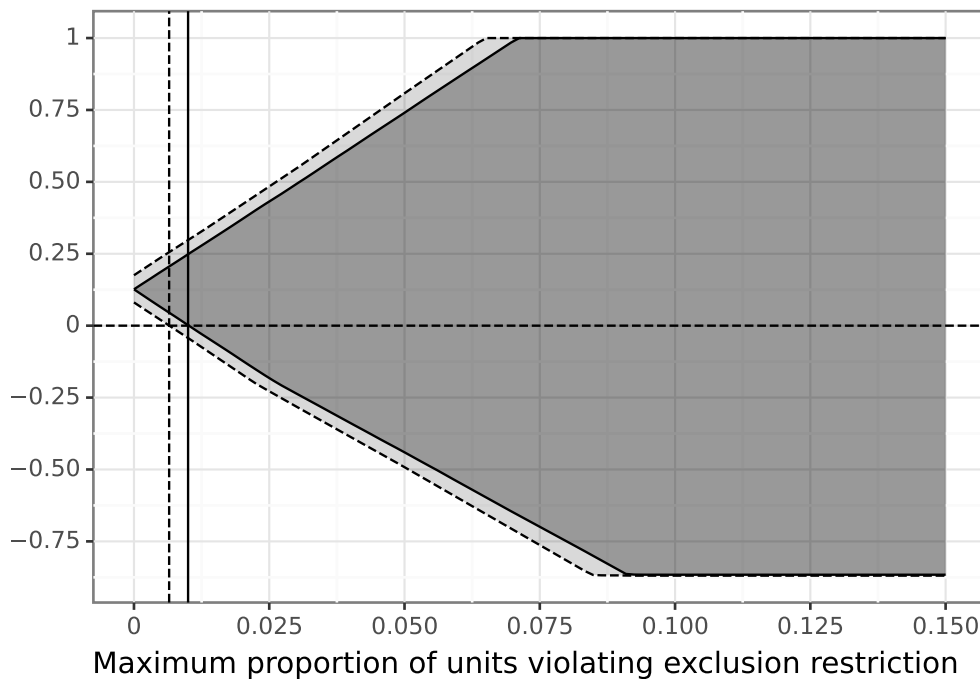


Figure 4: Sensitivity analysis of estimates of 95% confidence bounds of the LATE with respect to violations of exclusion restriction. The maximum proportion of units violating exclusion restriction (θ_{ER}) is represented on the horizontal axis. For this particular case, the bounds in the ATE do not change much with varying θ_{ER} . The bounds on LATE, however, widen from a point-identifiable estimate when θ_{ER} is 0 to large bounds when as θ_{ER} increases. Solid lines indicate the point estimates of the bounds, while the dashed lines represent the 2.5% and the 97.5% quantiles for respectively the lower and the upper bounds of the resamples. The solid vertical line in the graph indicates the value of $\theta_{\text{ER}} = 0.01$ for which the estimated lower bound crosses 0 for the point-estimate, while the dashed vertical line indicates the point where the confidence region crosses 0 at $\theta_{\text{ER}} = 0.0065$. In other words, if we have reason to believe the number of units violating exclusion restriction exceeds 1% of our sample for the point estimate and 0.65% for the confidence region, the sign of the estimated effect cannot be determined.



5.2 Classic Confounding: Reanalysis of [Hobolt \(2016\)](#)

To demonstrate how researchers might apply this method in a case of simple confounding, I replicate and extend [Hobolt \(2016\)](#), an investigation into the individual-level causes of support for “Brexit,” the landmark 2016 referendum in which British voters elected to break ties with the European Union. This study argues that “the divide between winners and losers of globalization was a key driver of the vote” (abstract). To test this claim, the study estimates various multivariate logistic regressions using individual-level survey data measuring support for Brexit (the dependent variable). While the study makes numerous causal claims, for illustrative purposes, we focus on the following statement: “Going from A-level education to an undergraduate degree reduces the probability of voting Leave by about 10 percentage points, all other things being equal” (1269). In other words, the study claims that voters’ level of education was a causal factor influencing the Brexit outcome.

It is important to note that, like many empirical social science studies, this paper does not formally state an estimand ([Lundberg et al., 2021](#)), making it difficult to evaluate. If a study never states its goal, it is impossible to determine whether the estimation strategy is appropriate. For the sake of demonstration, we assume here that the goal was to estimate the following average treatment effect (ATE):

$$\text{ATE}_{3 \rightarrow 4} = \mathbb{E}[Y(d = 4) - Y(d = 3)]$$

where $d = 4$ in this case indicates a voter having an undergraduate degree and $d = 3$ indicates a voter having an A-level education, following [Hobolt \(2016\)](#). For the quantity above to be reliably approximated with the estimation strategy in this paper, several assumptions must hold. First, the assumed functional form—a logistic regression—must be correct. Second, and most relevant for this discussion, conditional unconfoundedness must hold. That is, all factors X which jointly determine treatment D (education) and outcome Y (support for Brexit) are (i) measured and (ii) modeled correctly. According to Table 2 (Model 1) in the original paper, X includes gender, age, income, and a personal evaluation of the state of the economy. We note that these control variables have virtually no impact—regardless of whether they are

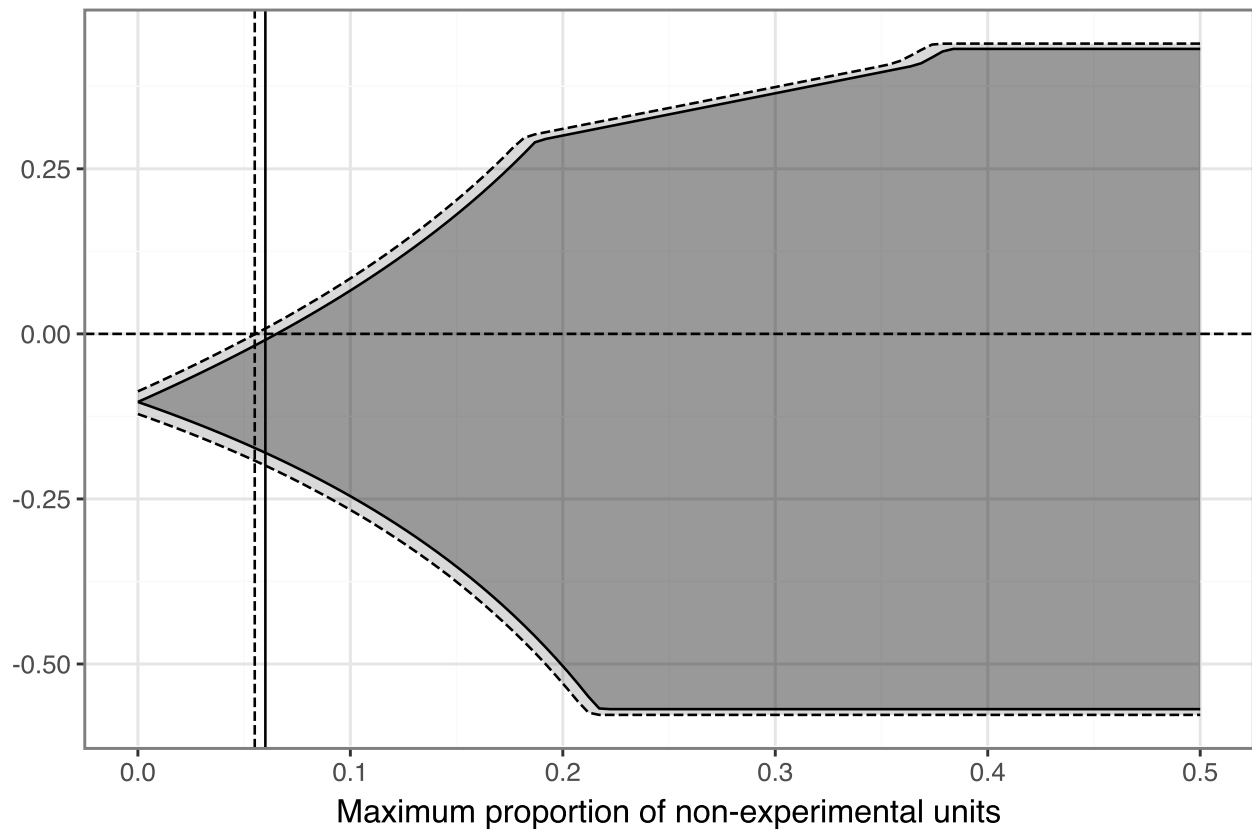
included or excluded, the estimated difference in support is -10 percentage points.

However, if the included control variables fail to account for all common causes of education and Brexit preference, then the ATE will necessarily be biased. This might be the case if there is a variable, such as living in a particular region or personal connections, impacting both taking an undergraduate degree and voting against Brexit simultaneously. Those factors are often unobservable, which is the case in this paper. What is the most we could learn about the $ATE_{3 \rightarrow 4}$ if we did not assume selection on observables? If confounding cannot be limited in any way, the best information one can get are bounds, which in the case of confounding are not often informative. In summary, if one assumes all confounders were included, then the estimate is -0.103 . On the other hand, if one believes there are unobservable confounders, such as network effects that cannot be adjusted for, then one can calculate bounds of $[-0.568, 0.432]$ with a 95% confidence region of $[-0.577, 0.44]$.

Fortunately, one can still employ sensitivity methods to evaluate how the result would change if we allowed for some confounding. Using the approach of [VanderWeele and Ding \(2017\)](#), an e-value of 1.90 might be obtained, and 1.78 if considering the confidence intervals. This e-value seems relatively weak. In fact, there is no general rule for interpreting this value to determine if it is small or large.

Using the approach advocated in this paper, one imagines there are two types of units. For one type, previously constructed personal connections make individuals more likely to obtain an undergraduate degree and vote against Brexit. For the other type, it might be the case that individuals obtained the degree, but the connections formed during that degree are the ones influencing them to vote against Brexit. The sensitivity curve is plotted in [Figure 5](#). Considering the point estimates, 6% of non-experimental units is sufficient to drive the effect away, while considering the 95% confidence region, this proportion falls to 5.5%. A relevant question is whether this value is high or low. Simulations suggest that values greater than 5% indicate robustness of results. In summary, the conclusion about the negative effect of education on voting in favor of Brexit is robust.

Figure 5: Sensitivity analysis of the ATE with respect to violations of unconfoundedness. The proportion of non-experimental (confounded) units (θ_{UC}) is represented on the x-axis. All bounds are estimated using bootstrap with 1000 resamples. Solid lines indicate the median of those resamples, while the dashed lines represent the 2.5% and 97.5% quantiles for the lower and upper bounds, respectively. The vertical line in the graph indicates the proportion at which the estimated upper bound crosses 0, which is 5.5%. In other words, if the proportion of non-experimental units is assumed to be above 5.5%, it means the estimated effect cannot be detected as positive.



5.3 More Complex Scenarios: Reanalysis of [Knox et al. \(2020\)](#)

In this section, I reanalyze results from [Knox et al. \(2020\)](#). The authors focus on estimating a particular effect of a civilian’s perceived race on whether a police officer uses force. They point out that administrative records, such as police reports, are often incomplete and biased because police only complete these reports for the selected subset of civilians that they choose to stop, rather than all civilians encountered; moreover, they point out that this initial decision to stop may itself be discriminatory. They then argue that by conducting regressions on selected stop data, past work has implicitly targeted the effect of perceived race in this subset of units defined on a post-treatment variable that is influenced by both observable and unobservable factors. Thus, if race or minority status affects both the decision to use force and the decision to stop individuals, analyses that do not account for this will be biased. The authors emphasize that traditional methods relying on documented encounters can significantly underestimate racial disparities, highlighting the need to account for selective reporting to more accurately estimate the ATE in documented encounters.

Knox, Lowe, and Mummolo make several critical assumptions in their analysis of racial bias in policing. One key assumption is *mandatory reporting*, which implies that force is not used on individuals who were not previously stopped. This assumption is generally reasonable and uncontested. However, other assumptions are based on qualitative knowledge of case history. For example, the assumption of *stop monotonicity* suggests that the probability of stopping an individual from a minority group is higher than that of stopping a white individual. This assumption supports their argument that even if some bias against white civilians exists, it does not explain the observed disparities, indicating a substantial underestimation of bias against minority civilians. These assumptions are central to their estimation strategy and underscore the potential limitations and biases of relying solely on administrative records to measure racial bias in policing. Although the *stop monotonicity* assumption appears reasonable, it is not guaranteed to hold in every case. In some scenarios, police officers might exhibit anti-white bias. This is where sensitivity analysis becomes relevant. By relaxing this assumption using the method outlined here, I find that the proportion of encounter cases

with anti-white bias must be less than 6.3%. This result is robust. Even if there is some anti-white behavior by police officers, to eliminate racial bias against minorities, one would need to assume a high level of anti-white bias, which seems counterintuitive.

6 Conclusion

In this paper, I propose a unified and automated method for sensitivity analysis, designed to work with a broad class of causal models. This approach offers a robust framework for assessing the impact of violations to causal assumptions, which can be categorized into three types: (a) functional assumptions, (b) assumptions of no direct effects, and (c) unconfoundedness assumptions. By calculating partial identification bounds based on the maximum proportion of units that violate these assumptions, researchers can evaluate not only how bounds shift with slight changes in this proportion but also determine the minimum proportion that drives a particular result away.

Through detailed case studies, such as habit formation in voting, analysis of racial bias in policing, and the effects of education on voting preferences, I demonstrate how this method can uncover important insights into the robustness of causal inferences. For example, applying the sensitivity analysis to Brexit voting behavior reveals the robustness of the effect of having an undergraduate degree on the likelihood of not voting to leave. Similarly, the analysis of police use of force highlights that even assuming a slight anti-white bias among officers does not sufficiently explain the existing bias against minorities. The automated nature of this method, implemented as a recent extension of the **autobounds** partial identification software suite, ensures its applicability to complex models and large datasets, making it a valuable tool for researchers across social sciences and other disciplines. By providing sharp bounds on causal effects and evaluating the influence of potential assumption violations, this method enhances the credibility and interpretability of causal estimates.

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A Principal Strata methods

A.1 Graph Stratification

Let an Acyclic Direct Mixed Graph with $G = (V, E, E_b)$. To create a stratified version of this graph, we use the procedure proposed by [Balke and Pearl \(1994\)](#):

Algorithm 1 ADMG stratification ([Balke and Pearl, 1994](#))

Input : Acyclic Direct Mixed Graph $G = (V, E, E_b)$

Output: Stratified set S_C

foreach $v \in V$ **do**

 | let S_v of all possible functions from $PA(v)$ to v with $\# \{S_v\} = \# \{v\}^{\prod_{u \in PA(v)} \# \{u\}}$

end

foreach $c \in C(V)$ **do**

 | let $S_c = S_{v_1} \times \dots \times S_{v_n}$ for all $v \in c$

end

return $S_C = \{S_c, \forall c \in C(V)\}$

The procedure described in Algorithm 1 makes use of the idea of c-component decomposition ([Tian, 2002](#); [Shpitser and Pearl, 2006](#)), the sets of variables connected by bidirected edges. Notice, however, that there are extra constraints that are not captured by the procedure. For instance, in an instrumental variable model where the instrument Z is connected with D , and D with Y through bidirected edges, but there is no bidirected edge between Z and Y , this extra constraint is added through:

Algorithm 2 Extra constraints

Input : Acyclic Direct Mixed Graph $G = (V, E, E_b)$, Stratified Graph S_C

Output: C-component extra constraints X_C

foreach $v \in V$ **do**

 | let S_v of all possible functions from $PA(v)$ to v with $\# \{S_v\} = \# \{v\}^{\prod_{u \in PA(v)} \# \{u\}}$

end

foreach $c \in C(V)$ **do**

foreach $v, u \in c$ s.t. *bidirected edge* v, u not in E_b **do**

 Let $S_c \in S_C$

 Let $X_{v,u} = \left\{ \left(\sum_{k \in S_c | u_i, v_j \in k} P(k) - \sum_{l \in S_c | u_i \in l} P(l) \cdot \sum_{m \in S_c | v_j \in m} P(m) \right) = 0 \right\}$

$\forall u_i \in u, v_j \in v$

end

end

The ordered pair $C = (S_C, X_C)$ generated by algorithms 1 and 2 generate the canonical-ization of G , which will be used in the optimization partial identification procedure. Only S_C is required, however, to translate all the other constraints from G .

A.2 Principal strata for instrumental variable models

Notice that the strata for Z are z_0 and z_1 and are immediately identifiable. Then we stratified A and Y , which are confounded as,

$$a_{00}y_{00}, a_{00}y_{01}, a_{00}y_{10}, a_{00}y_{11}, a_{01}y_{00}, \dots, a_{11}y_{11}$$

B Sensitivity Function Results

B.1 Symbolic sensitivity function for simple confounding

Suppose one has a model with binary treatment A and outcome Y and both are confounded. Assume also that one know θ_{UC} , which is the proportion of non-experimental units in the population. To analyze this scenario, we do first the canonicalization.

We reconstruct first each piece of data:

$$\begin{aligned} P(A = 1, Y = 1) &= \theta_{UC}(a_1y_{01} + a_1y_{11}) + (1 - \theta_{UC})(a_1 \cdot (y_{01} + y_{11})) \\ P(A = 1, Y = 0) &= \theta_{UC}(a_1y_{10} + a_1y_{00}) + (1 - \theta_{UC})(a_1 \cdot (y_{10} + y_{00})) \\ P(A = 0, Y = 1) &= \theta_{UC}(a_0y_{10} + a_0y_{11}) + (1 - \theta_{UC})(a_0 \cdot (y_{10} + y_{11})) \\ P(A = 0, Y = 0) &= \theta_{UC}(a_0y_{01} + a_0y_{00}) + (1 - \theta_{UC})(a_0 \cdot (y_{01} + y_{00})) \end{aligned}$$

We also reconstruct a simple estimand $P(Y(A = 1) = 1)$:

$$P(Y(A = 1) = 1) = \theta_{UC}(a_1y_{01} + a_1y_{11} + a_0y_{01} + a_0y_{11}) + (1 - \theta_{UC})(y_{01} + y_{11})$$

The extension is straightforward to $P(Y(A = 0) = 1)$, the ATE, or other estimands.

Solving this program using autobounds or other polynomial program solvers can yield sharp bounds for the estimands of interest.

To obtain intuition on the problem, we derive closed-form bounds as well. Despite not proving their sharpness, tests using autobounds provide evidence that they are indeed the sharp closed-form bounds.

Lemma 2. *In the scenario described in this subsection, sharp bounds for $y_{01} + y_{11}$ are given by:*

$$LB(y_{01} + y_{11}) = \begin{cases} \frac{P(A=1, Y=1) - \theta_{UC}}{P(A=1) - \theta_{UC}}, & \text{if } \frac{P(A=1, Y=1) - \theta_{UC}}{P(A=1) - \theta_{UC}} \in [0, 1] \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

and

$$UB(y_{01} + y_{11}) = \begin{cases} \frac{P(A=1, Y=1)}{P(A=1) - \theta_{UC}}, & \text{if } \frac{P(A=1, Y=1)}{P(A=1) - \theta_{UC}} \in [0, 1] \\ 1, & \text{otherwise} \end{cases} \quad (6)$$

Proof. To construct the upper bound for $y_{01} + y_{11}$, it is sufficient to calculate bounds for $a_1 \cdot (y_{01} + y_{11})$ and for a_1 . Then $UB(y_{01} + y_{11}) = UB(a_1 \cdot (y_{01} + y_{11})) / LB(a_1)$. The upper bound for $a_1 \cdot (y_{01} + y_{11}) = (P(A = 1, Y = 1) - \theta_{UC}(a_1y_{01} + a_1y_{11})) / (1 - \theta_{UC})$. Assuming that the second term attains its lowest value, 0, then the upper bound will be $P(A = 1, Y = 1) / (1 - \theta_{UC})$. To calculate the lower bound for a_1 , we calculate first the upper bound for a_0 , which will be equal to $P(A = 0) / (1 - \theta_{UC})$. Then the $LB(a_1) = 1 - P(A = 0) / (1 - \theta_{UC}) = (P(A =$

$1) - \theta_{UC}) / (1 - \theta_{UC})$. The upper bound for $y_{01} + y_{11}$ is , so, $P(A = 1, Y = 1) / (P(A = 1) - \theta_{UC})$.

To calculate the lower bound for $y_{01} + y_{11}$, we use an analogous reasoning for the upper bound of $y_{10} + y_{00}$. Then $LB(y_{01} + y_{11}) = 1 - UB(y_{10} + y_{00}) = 1 - P(A = 1, Y = 0) / (P(A = 1) - \theta_{UC}) = (P(A = 1, Y = 1) - \theta_{UC}) / (P(A = 1) - \theta_{UC})$.

Notice that this formula might return bounds that are outside the probability region of $[0, 1]$. For instance, if $P(A = 1) = \theta_{UC}$, then the denominator of both lower and upper bound will be 0, yielding an invalid results. In such invalid cases, the lower bound will be the lowest as possible, 0, and the upper bound, the greatest as possible, 1.

□

Lemma 3. *In the scenario described in this subsection, sharp bounds for $a_1y_{01} + a_1y_{11} + a_1y_{01} + a_1y_{11}$ are given by:*

$$LB(a_1y_{01} + a_1y_{11} + a_1y_{01} + a_1y_{11}) = \min\{1, (1 - P(A = 1, Y = 1)) / \theta_{UC}\} \quad (7)$$

and

$$UB(a_1y_{01} + a_1y_{11} + a_1y_{01} + a_1y_{11}) = \max\{0, (\theta_{UC} - P(A = 1, Y = 1)) / \theta_{UC}\} \quad (8)$$

Proof. Notice that $P(A = 1, Y = 1) \leq P(Y(A = 1) = 1)$ is a valid lower bound and symmetrically $1 - P(A = 1, Y = 0)$ is a valid upper bound as $1 - P(A = 1, Y = 0) \geq 1 - P(Y(A = 1) = 0) = P(Y(A = 1) = 1)$. Those are sharp bounds for the total confounded case, i.e. $\theta_{UC} = 1$.

To reconstruct the bounds for the estimand, we firstly construct bounds only for $a_1y_{01} + a_1y_{11} + a_0y_{01} + a_0y_{11}$, which is the confounded part of the model. To calculate the upper bound for it, we firstly assume that the estimand $P(Y(A = 1) = 1)$ attains its upper bound at $1 - P(A = 1, Y = 0)$ and the unconfounded part, $y_{01} + y_{11}$ is the lowest as possible, 0. Then

$$\begin{aligned} \theta_{UC}(a_1y_{01} + a_1y_{11} + a_0y_{01} + a_0y_{11}) + (1 - \theta_{UC}) \cdot 0 &\leq 1 - P(A = 1, Y = 1) \\ a_1y_{01} + a_1y_{11} + a_0y_{01} + a_0y_{11} &\leq (1 - P(A = 1, Y = 1)) / \theta_{UC} \end{aligned}$$

The lower bound can be calculated using an analogous reasoning, assuming the worst-case of $y_{01} + y_{11}$ being the greatest as possible, 1. Then

$$a_1y_{01} + a_1y_{11} + a_0y_{01} + a_0y_{11} \geq (\theta_{UC} - P(A = 1, Y = 1)) / \theta_{UC}$$

Finally, notice that the confounded part cannot be greater than 1 or lower than 0, then:

$$\min\{1, (1 - P(A = 1, Y = 1)) / \theta_{UC}\} \leq a_1y_{01} + a_1y_{11} + a_0y_{01} + a_0y_{11} \leq \max\{0, (\theta_{UC} - P(A = 1, Y = 1)) / \theta_{UC}\}$$

□

Theorem 4 (Sensitivity bounds under classic confounding). *A valid sensitivity function for $P(Y(A = a) = y)$ in the classic confounding scenario is*

$$\min\{\theta_{UC} \cdot \Delta_{lb}^1 + (1 - \theta_{UC}) \cdot \Delta_{lb}^2, \Delta_{lb}^3\} \leq P(Y(A = a) = y) \leq \max\{\theta_{UC} \cdot \Delta_{ub}^1 + (1 - \theta_{UC}) \cdot \Delta_{ub}^2, \Delta_{ub}^3\} \quad (9)$$

where

$$\Delta_{lb}^1 = \max\{1, \frac{P(Y = y, A = a)}{\theta_{UC}}\},$$

$$\Delta_{ub}^1 = \min\{0, 1 - \frac{P(Y = 1 - y, A = a)}{\theta_{UC}}\},$$

$$\Delta_{lb}^2 = \begin{cases} \frac{P(A=a, Y=y) - \theta_{UC}}{P(A=a) - \theta_{UC}}, & \text{if } \frac{P(A=a, Y=y) - \theta_{UC}}{P(A=a) - \theta_{UC}} \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$\Delta_{ub}^2 = \begin{cases} \frac{P(A=a, Y=y)}{P(A=a) - \theta_{UC}}, & \text{if } \frac{P(A=a, Y=y)}{P(A=a) - \theta_{UC}} \in [0, 1] \\ 1, & \text{otherwise} \end{cases}$$

$$\Delta_{lb}^3 = \max\{(1 - \theta_{UC}) \cdot P(Y = y|A = a), P(Y = y, A = a)\},$$

and

$$\Delta_{ub}^3 = \min\{\theta_{UC} + (1 - \theta_{UC}) \cdot P(Y = y|A = a), 1 - P(Y = 1 - y, A = a)\}$$

Proof. Closed-form solutions for Δ_{lb}^2 and Δ_{ub}^2 can be obtained directly from Lemma 2. Closed-form solutions for Δ_{lb}^2 and Δ_{ub}^2 can be obtained by using a reasoning similar to the one employed in Lemma 3, inverting lower and upper bound. That entails the bounds $\theta_{UC} \cdot \Delta_{lb}^1 + (1 - \theta_{UC}) \cdot \Delta_{lb}^2$.

However, these lower (upper) bounds are not sufficient because whenever they are greater (lower) than Δ_{lb}^3 (Δ_{ub}^3), which is parameterized by the convex combination of identifiable point when θ_{UC} is 0 and 0 (1) when θ_{UC} is 1, they will not be valid bounds. Then it is necessary to collect the minimum (maximum) of both. $\square$

Remark 1. To construct bounds for the ATE, risk ratio, and other estimands that depend on $P(Y(A = 1) = 1)$ and $P(Y(A = 0) = 1)$, we might combine them. For instance, the lower bound for the ATE, $P(Y(A = 1) = 1) - P(Y(A = 0) = 1)$, is $LB(P(Y(A = 1) = 1)) - UB(P(Y(A = 0) = 1))$.

B.2 Symbolic sensitivity function for “no defiers” in instrumental variable models

I reconstruct each piece of data conditioned on Z (notice that $P(Z = z)$ is immediately identifiable).

$$\begin{aligned}
\Pr(y_0, a_0 | z_0) &= a_{00}y_{00} + a_{01}y_{00} + a_{00}y_{01} + a_{01}y_{01}, \\
\Pr(y_0, a_0 | z_1) &= a_{00}y_{00} + a_{10}y_{00} + a_{00}y_{01} + a_{10}y_{01}, \\
\Pr(y_0, a_1 | z_0) &= a_{11}y_{00} + a_{10}y_{00} + a_{11}y_{10} + a_{10}y_{10}, \\
\Pr(y_0, a_1 | z_1) &= a_{11}y_{00} + a_{01}y_{00} + a_{11}y_{10} + a_{01}y_{10}, \\
\Pr(y_1, a_0 | z_0) &= a_{00}y_{11} + a_{01}y_{11} + a_{00}y_{10} + a_{01}y_{10}, \\
\Pr(y_1, a_0 | z_1) &= a_{00}y_{11} + a_{10}y_{11} + a_{00}y_{10} + a_{10}y_{10}, \\
\Pr(y_1, a_1 | z_0) &= a_{11}y_{11} + a_{10}y_{11} + a_{11}y_{01} + a_{10}y_{01}, \\
\Pr(y_1, a_1 | z_1) &= a_{11}y_{11} + a_{01}y_{11} + a_{11}y_{01} + a_{01}y_{01},
\end{aligned}$$

I also add the sensitivity piece, the probability of defiers: $\theta_1 = a_{10}y_{00} + a_{10}y_{01} + a_{10}y_{10} + a_{10}y_{11}$.

Finally, the ATE can be written as:

$$ATE = a_{00}y_{01} + a_{01}y_{01} + a_{10}y_{01} + a_{11}y_{01} - a_{00}y_{10} - a_{01}y_{10} - a_{10}y_{10} - a_{11}y_{10}$$

Solving this optimization problem assuming the ATE is the objective function while the other pieces are constraints is sufficient to obtain the closed-form solutions.

Bounds for LATE can be found by using the estimand as:

$$LATE = \frac{a_{01}y_{01} - a_{01}y_{10}}{a_{01}y_{00} + a_{01}y_{01} + a_{01}y_{10} + a_{01}y_{11}}$$

To calculate bounds for the case of exclusion restriction, one just has to re-derive the problem with the case where Y has two parents (direct effect from Z and A on Y), so there will be 16 strata $y_{0000}, \dots, y_{1111}$. The parameter for exclusion restriction violation are those where changing Z affects the effect of A on Y :

$$\theta_2 = y_{0001} + y_{0010} + y_{0100} + y_{0110} + y_{0111} + y_{1000} + y_{1001} + y_{1011} + y_{1101} + y_{1110}$$